

To my family.

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Disclosure on the Use of Artificial Intelligence

In preparing this thesis the author used three large language models — Anthropic’s Claude ([Anthropic, 2025](#)), OpenAI’s ChatGPT ([OpenAI, 2025](#)) and Google’s Gemini ([Google DeepMind, 2025](#)) — as assistive tools, in three capacities: writing, refactoring and documenting the analysis, experiment-management and figure-generation code of the research repository, including the $\text{\LaTeX}$ build and consistency checks; reading drafts to flag possible mathematical slips, inconsistencies between text, tables and figures, and unclear passages; and editing prose for clarity and assisting with $\text{\LaTeX}$ typesetting.

The research questions, the mathematical derivations, the experimental design, the interpretation of the results and the conclusions are the author’s, every AI-assisted change was reviewed by the author before inclusion, and the author takes full responsibility for the content. No number, table entry or figure in this document was produced by a language model: all reported quantities are computed by the version-controlled pipeline of [Appendix A](#) from the frozen experiment outputs it documents, and the figures and generated tables are built by the scripts that appendix lists.

Chapter 1

Introduction

1.1 Motivation and gap

Generative diffusion models dominate the synthesis of high-dimensional data (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021): a forward process noises the data, a learned reverse process rebuilds it, and the one object the reverse process needs is the *score* $S(x, t) = \nabla_x \log p_t(x)$ of the noised law (Anderson, 1982). In practice the score is a network trained by denoising score matching (Vincent, 2011), and for essentially every data law of interest the score it should have produced is unavailable (Kadkhodaie et al., 2024). Exactly solvable models supply such a reference (Biroli and Mézard, 2023; Biroli et al., 2024), but treat data as unstructured samples, so the internal time of a *sequence* does not appear in them, and the classical inference literature on Gauss–Markov chains (Kalman, 1960; Mézard and Montanari, 2009) does not ask how the exact posterior, hence the exact score, deforms as the noise is swept from zero to saturation. This thesis works in that gap, on sequences generated by a first-order Markov chain and corrupted coordinatewise by the Ornstein–Uhlenbeck channel. The corruption adds no edges among the latent variables, so the posterior factor graph is a tree on which sum–product is exact, and the *joint* score of the whole noisy sequence, the object one-frame scores are blind to (Chapter 3), is an exact recursion even for non-Gaussian innovations. The thesis computes it exactly where a closed form exists, approximates it in a measured way where none does, and learns it from noised data when the dynamics are unknown.

1.2 Research questions

- RQ1.** For a stationary Gaussian AR(1) chain under coordinatewise OU noise, what is the joint score, and how do its couplings vary with diffusion time?
- RQ2.** How is the same score computed by belief propagation, and at what cost?
- RQ3.** For non-Gaussian innovations, how accurate is a Gaussian-message approximation against a validated numerical reference?
- RQ4.** How fast does the influence of remote observations on the score decay, and what does this say about local approximations?
- RQ5.** Under what assumptions can an *unknown* transition law be recovered from noised sequences alone, and which checks make that reliable?
- RQ6.** Where the Markov assumption holds exactly, what is supplying it worth against networks trained on the same data and given none of it?

1.3 Contributions and structure

1. **The Gaussian case worked out completely:** closed-form score, spectrum and coupling lifecycle, chain belief propagation identified with Kalman filtering and RTS smoothing, and the measured price of ignoring distant evidence (Chapters 3–4; RQ1, RQ2, RQ4).
2. **What Gaussian closure computes off the Gaussian case:** the LMMSE estimator of the covariance-matched model (Theorem 5.2), which makes the measured closure gap the price of discarding structure beyond second moments (Chapter 5; RQ3).
3. **Learning the chain, and what structure is worth:** an unknown kernel fitted by EM with Fisher’s identity supplying the exact gradient from one forward–backward pass, its two convergence rates, and a measured value of $6.6\text{--}10.9\times$ against an unstructured network and $2.3\text{--}6.1\times$ against one given locality, not a ranking of algorithms (Chapters 6–8; RQ5, RQ6).

Chapter 2 reviews the literature; Chapter 9 answers the six questions.

Chapter 2

Statistical Mechanics, Energy-Based Models, and Diffusion

This chapter reads generative diffusion through the statistical-mechanics lineage it came from, fixes the score–posterior identity that every later chapter uses, and locates the gap the thesis addresses; textbook material is stated and cited, not derived.

2.1 From the Boltzmann law to energy-based models

Statistical mechanics abandons the trajectory of a many-body system and keeps a probability law over its microstates. Among all laws with a given mean energy, the equilibrium one maximises the entropy $S[p] = -\sum_x p(x) \log p(x)$ (Jaynes, 1957), and the maximiser is the Boltzmann distribution

$$p(x) = \frac{e^{-\beta E(x)}}{Z(\beta)}, \quad Z(\beta) = \sum_x e^{-\beta E(x)}, \quad (2.1)$$

with the inverse temperature β as the Lagrange multiplier of the energy constraint (Landau and Lifshitz, 1980). Two features of (2.1) recur throughout. The macroscopic behaviour is encoded in the single scalar $\log Z$, a sum over exponentially many configurations that is #P-hard for broad classes of interacting models (Barahona, 1982; Jerrum and Sinclair, 1993); the tractable exceptions, trees and Gaussians, are exactly the structures later chapters exploit. And the competition between energy and entropy

can produce *phase transitions*, changes of character of the equilibrium state as a parameter is varied without any change in the microscopic law (Ising, 1925; Onsager, 1944).

Energy-based models read (2.1) in the opposite direction, in a lineage that runs from the Hopfield network and its spin-glass capacity analysis (Hopfield, 1982; Amit et al., 1985; Mézard et al., 1987) to the Boltzmann machine (Ackley et al., 1985) and the 2024 Nobel Prize in Physics (The Royal Swedish Academy of Sciences, 2024): one posits a parametric energy E_θ and fits it so that $p_\theta(x) = e^{-E_\theta(x)}/Z(\theta)$ matches data (LeCun et al., 2006). The maximum-likelihood gradient is a competition between two phases,

$$\nabla_\theta \mathbb{E}_{x \sim p_{\text{data}}} [-\log p_\theta(x)] = \mathbb{E}_{x^+ \sim p_{\text{data}}} [\nabla_\theta E_\theta(x^+)] - \mathbb{E}_{x^- \sim p_\theta} [\nabla_\theta E_\theta(x^-)], \quad (2.2)$$

which follows from $\nabla_\theta \log Z(\theta) = -\mathbb{E}_{p_\theta} [\nabla_\theta E_\theta]$: the positive phase lowers the energy of data, the negative phase raises the energy of the model's own samples. The second expectation is over p_θ , which cannot be sampled directly; approximating it by Markov chain Monte Carlo (Hinton, 2002; Robert and Casella, 2004) is what made the family expensive and, for a long time, outperformed by models that give up the explicit density for cheap sampling (Goodfellow et al., 2014; Kingma and Welling, 2014; Papamakarios et al., 2021). The training dynamics of these models is itself a statistical-mechanics problem: Bachtis et al. (2024) show that a restricted Boltzmann machine learns its features through a cascade of second-order transitions, coarsest first, an ordering that Section 2.4 meets again along the generation trajectory.

2.2 Generative diffusion

A diffusion model destroys structure in a controlled way and learns to run the destruction backwards (Sohl-Dickstein et al., 2015; Song and Ermon, 2019; Ho et al., 2020; Song et al., 2021); its deterministic counterparts, neural ODEs and flow matching, transport the same marginals along a deterministic path (Chen et al., 2018; Lipman et al., 2023).

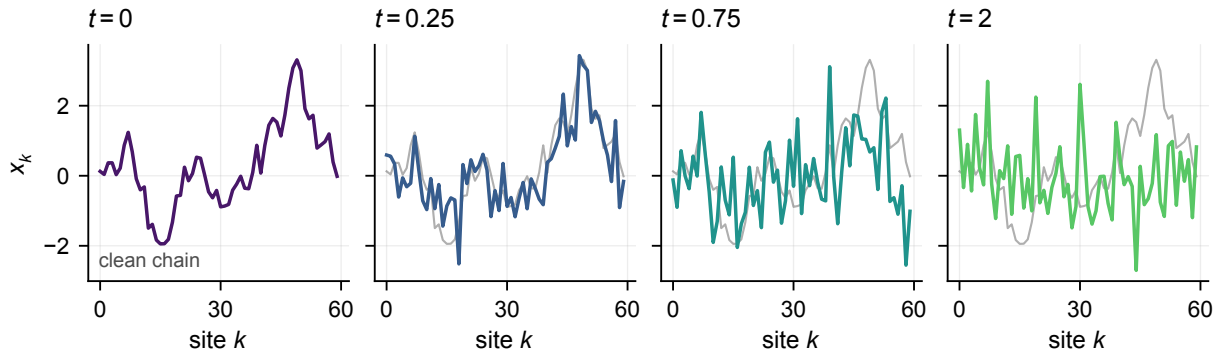


Figure 2.1: One clean stationary AR(1) trajectory with $\alpha = 0.85$ (grey) and the same trajectory after the OU channel (2.4) at four diffusion times. The signal contracts by e^{-t} while independent noise of variance Δ_t accumulates; correlation between neighbouring sites outlives the original amplitude. Generation runs this evolution from right to left.

The forward process is the SDE

$$dX_t = f(X_t, t) dt + g(t) dW_t, \quad X_0 \sim p_0, \quad (2.3)$$

whose marginals p_t obey the Fokker–Planck equation $\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g^2 \Delta p_t$ (Gardiner, 2009). The choice used throughout this thesis is not arbitrary: overdamped Langevin dynamics in a confining potential E has the Boltzmann law $\propto e^{-\beta E}$ as its unique stationary density, and $\beta = 1$, $E(x) = \frac{1}{2} \|x\|^2$ gives $f = -x$, $g = \sqrt{2}$, the Ornstein–Uhlenbeck (OU) channel (Uhlenbeck and Ornstein, 1930), whose transition law is available in closed form,

$$x = e^{-t} a + \sqrt{\Delta_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad \Delta_t = 1 - e^{-2t}. \quad (2.4)$$

The signal contracts by e^{-t} while isotropic noise of variance Δ_t fills in; the terminal law is the equilibrium of a quadratic Hamiltonian at unit temperature, and the forward process is a relaxation to it from the data law (Figure 2.1). The original construction of Sohl-Dickstein et al. (2015) took exactly this non-equilibrium viewpoint: rather than normalising a global energy, it compares forward and learned reverse *path* measures, and the resulting evidence lower bound is tractable because every forward transition is known and locally normalised.

What determines the reverse dynamics is a classical result. For a state-independent $g(t)$ and smooth, positive marginals with integrable score, [Anderson \(1982\)](#) shows that

$$dX_t = [f(X_t, t) - g(t)^2 \nabla_x \log p_t(X_t)] dt + g(t) d\bar{W}_t, \quad (2.5)$$

integrated from T back to 0 from $X_T \sim p_T$, reproduces the forward marginals in reverse order; the probability-flow ODE $\dot{X}_t = f - \frac{1}{2}g^2 \nabla_x \log p_t$ has the same one-time marginals with no Brownian term ([Song et al., 2021](#)). Both are driven by the same object, the score of the noised law,

$$s_t(x) := \nabla_x \log p_t(x), \quad (2.6)$$

and so is every practical parameterisation: predicting the injected noise, as in [Ho et al. \(2020\)](#), is the same conditional expectation rescaled ([Lai et al., 2025](#)).

2.3 The score–posterior identity, and three scores

The word “score” is used for three different objects, and almost every difficulty in generative diffusion is a property of the gaps between them ([Biroli and Mézard, 2023](#)).

The exact score. The score of the true noised law $p_t = (\text{contracted } p_0) * \mathcal{N}(0, \Delta_t I)$. Writing $p_t(x) = \int p_0(a) G(x; e^{-t}a, \Delta_t) da$ with G the Gaussian kernel of (2.4), differentiating under the integral and using $\nabla_x G = -\Delta_t^{-1}(x - e^{-t}a) G$ gives

$$\boxed{\nabla_x \log p_t(x) = \frac{e^{-t} \mathbb{E}[a | x] - x}{\Delta_t} \iff \mathbb{E}[a | x] = e^t (x + \Delta_t \nabla_x \log p_t(x))}, \quad (2.7)$$

where $\mathbb{E}[a | x]$ is the mean of the posterior $p_t(a | x) \propto p_0(a) G(x; e^{-t}a, \Delta_t)$, whose normaliser is $p_t(x)$ itself. The identity needs only $\int \|a\| p_0(a) da < \infty$. It is Tweedie’s formula of empirical Bayes ([Robbins, 1956](#); [Miyasawa, 1961](#); [Efron, 2011](#)), and it says more than an analogy: computing a score *is* solving a Bayesian denoising problem. For vector-valued data $a \in \mathbb{R}^L$ noised coordinatewise it holds componentwise but with the

full posterior,

$$(\nabla_x \log p_t(x))_k = \frac{e^{-t} \mathbb{E}[a_k \mid x_0, \dots, x_{L-1}] - x_k}{\Delta_t}, \quad (2.8)$$

because the channel is independent across frames but the prior is not. This is the exact sense in which the joint score of a trajectory differs from a stack of per-frame scores, and the reason this thesis studies the former.

The empirical score. Replacing p_0 by the empirical measure of a training set $\{a^{(i)}\}_{i=1}^N$ makes p_t a finite Gaussian mixture and the score a softmax-weighted average of training points,

$$\nabla_x \log p_t^{\text{emp}}(x) = \frac{1}{\Delta_t} \left(e^{-t} \sum_{i=1}^N w_i(x) a^{(i)} - x \right), \quad w_i(x) \propto e^{-\|x - e^{-t} a^{(i)}\|^2 / 2\Delta_t}. \quad (2.9)$$

It is exactly computable and exactly useless as a generative model: as $t \rightarrow 0$ the weights concentrate on the nearest training example and the reverse dynamics deposit the trajectory on top of it.

The fitted score. A network s_ϕ trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011) minimises

$$\mathbb{E}_t \mathbb{E}_{a \sim p_{\text{data}}} \mathbb{E}_\varepsilon \left\| s_\phi(e^{-t}a + \sqrt{\Delta_t} \varepsilon, t) + \frac{\varepsilon}{\sqrt{\Delta_t}} \right\|^2. \quad (2.10)$$

At the population level its minimiser over all measurable functions is the exact score; conditional on a finite training set it is the empirical score (2.9). A trained network reaches neither, and one proposed mechanism of generalisation is precisely this gap: the network is too constrained to represent (2.9), and that failure is what produces novel samples (Kadkhodaie et al., 2024; Kamb and Ganguli, 2024). On that account generalisation is an approximation-theoretic property of an architecture, a considerably less comfortable foundation than the variational principle behind (2.1). Deciding what a trained network has learned therefore requires the score it should have produced, and

for essentially every data law of interest that reference is unavailable. The solvable models of this thesis supply it.

2.4 Transitions along the generation trajectory

Because generation is a dynamical process, its failures are dated by two crossover times. *Speciation*: run backwards from noise, the trajectory commits to a broad class of outcomes long before it commits to a sample. In the Gaussian-mixture models of [Biroli et al. \(2024\)](#) and [Achilli et al. \(2026\)](#), for data whose covariance has top eigenvalue λ , commitment happens near $t_S \sim \frac{1}{2} \log \lambda$, when the contracted signal along that direction becomes comparable with the accumulated noise; for hierarchical data there is a ladder of such times, one per level, and features are resolved coarse to fine ([Sclocchi et al., 2025b,a](#); [Favero et al., 2025a,b](#)), the ordering [Bachtis et al. \(2024\)](#) observe in training. *Collapse and memorisation*: later in the reverse trajectory, dynamics driven by the empirical score are captured by a single training example past a time t_C , and avoiding this needs a training set exponentially large in the data dimension ([Biroli et al., 2024](#); [Bonnaire et al., 2025](#)); the criterion is entropic, collapse occurring when the excess entropy of the noised law over the terminal measure falls below $\log N$. Memorisation need not be a collapse onto one example ([Achilli et al., 2024](#)), the resulting generalisation can be biased ([Garnier-Brun et al., 2026](#)), and solvable models of the learning problem itself complete the picture ([Cui et al., 2025](#)). The two transitions bracket the useful part of the trajectory. The solvable-model analyses in this line treat a data point as an unstructured vector; the sequence axis of this thesis is absent from them.

2.5 Locality, architecture, and message passing

If the fitted score is doing the work, the question is what the architecture contributes. The network in almost every image or video diffusion model is a U-Net ([Ronneberger et al., 2015](#)): a convolutional encoder that downsamples, a decoder that upsamples, and long skip connections joining stages of equal resolution. [Mei \(2024\)](#) gives that

design a precise reading. For data generated by a tree-structured hierarchical model the exact denoiser is computed by belief propagation, an upward pass that accumulates evidence from the leaves and a downward pass that redistributes it, reusing the upward messages; a U-Net has exactly this shape, encoder, decoder and skips, so for those models it is an approximate implementation of the exact inference algorithm, with sample-complexity bounds to match. Nothing in that reading certifies an arbitrary U-Net on arbitrary data as message passing, and the locality that convolutional architectures build in (LeCun et al., 2015) is a statement about receptive fields, not about the probabilistic structure of the data.

Two consequences shape this thesis. Sum–product is exact on trees (Theorem 4.1), so on a chain the posterior marginals, and by (2.8) the joint score, are computed with no approximation at the inference stage: the structured reference against which learned denoisers are compared. And the same formulation makes the receptive field a quantity rather than a hyperparameter: the influence of an observation d sites away decays at a rate the model fixes, which Chapter 8 computes in closed form and uses to predict the best window radius before any network is trained. On sequences, Garnier-Brun et al. (2024) use exact belief propagation in the same role, as the reference for a transformer trained on hierarchically generated data, and find that correlations are learned shortest scale first.

2.6 Positioning and gap

Every generative family turns the same goal, drawing from a law p_0 nobody can write down, into a map from a trivially sampled reference law to samples: adversarial networks by a push-forward trained in a minimax game (Goodfellow et al., 2014), variational autoencoders by a stochastic decoder trained on a likelihood bound (Kingma and Welling, 2014), normalizing flows by an invertible map with exact likelihood (Papamakarios et al., 2021), autoregressive models one coordinate at a time (Vaswani et al., 2017), and diffusion by repeated denoising, sequential in the noise level rather than

in the dimension; [Ghio et al. \(2024\)](#) place flows, diffusion and autoregressive networks under one spin-glass analysis and find the same first-order transition governing sample quality in all three.

Three literatures meet in this thesis and leave a gap between them. The statistical physics of diffusion has closed-form analyses of the backward dynamics and its transitions, but for data without an internal time axis. Exact inference on Gauss–Markov chains is classical, through the Kalman filter and smoother and their reading as sum–product on a tree ([Kalman, 1960](#); [Rauch et al., 1965](#); [Pearl, 1988](#); [Kschischang et al., 2001](#); [Mézard and Montanari, 2009](#)), but it does not ask how the exact posterior, and so the exact score, deforms as the noise level is swept from zero to saturation. And message passing with continuous variables closes its messages inside a parametric family, the step connecting belief propagation to AMP and to the TAP equations ([Thouless et al., 1977](#); [Donoho et al., 2009](#); [Zdeborová and Krzakala, 2016](#)), on a central-limit argument that needs many messages per node and fails by construction on a chain. What follows computes the joint score of a noised Markov sequence exactly where that is possible, measures the price of Gaussian closure where it is not, and learns the dynamics from noised data when they are unknown, against a reference that is deterministic quadrature rather than Monte Carlo, because a discretisation error can be bounded separately and reported whereas a sampling error would enter every comparison.

Chapter 3

The Gaussian Chain by Linear Algebra

The stationary Gaussian AR(1) chain is the simplest non-trivial exactly solvable model in which temporal dependence survives Ornstein–Uhlenbeck corruption and the full joint score can be studied analytically. This chapter does that with matrices: the clean law, its dense covariance and sparse precision, the score in two equivalent forms, and the lifecycle the coupling structure passes through as the diffusion time grows. Chapter 4 obtains the same score by message passing.

3.1 The model and the object of study

A sequence $(a_k)_{k \geq 0}$ is a time-homogeneous Markov chain with transition kernel $K(a' \mid a)$ if the joint law of its first L states factorises into local terms,

$$p(a_0, \dots, a_{L-1}) = p_0(a_0) \prod_{k=0}^{L-2} K(a_{k+1} \mid a_k). \quad (3.1)$$

This is the most heavily used equation of the thesis: each factor touches two consecutive variables, which makes the chain a tree in the factor-graph sense of Chapter 4 and its precision tridiagonal in the Gaussian case below. The running example is the scalar Gaussian autoregression of order one (Brockwell and Davis, 1991),

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad \sigma_\eta^2 = 1 - \alpha^2, \quad a_0 \sim \mathcal{N}(0, 1), \quad |\alpha| < 1, \quad (3.2)$$

with kernel $K(a' | a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$. Since $\text{Var}(a_{k+1}) = \alpha^2 \text{Var}(a_k) + \sigma_\eta^2$, the choice $\sigma_\eta^2 = 1 - \alpha^2$ holds the marginal variance at one, so $a_k \sim \mathcal{N}(0, 1)$ for every k ; and unrolling the recursion, $a_{k+d} = \alpha^d a_k + (\text{innovations independent of } a_k)$, gives

$$(\Sigma_0)_{ij} := \text{Cov}(a_i, a_j) = \alpha^{|i-j|}. \quad (3.3)$$

Because a_0 is drawn from the fixed point of K , the chain is strictly stationary, and Σ_0 is symmetric Toeplitz: dense for $\alpha \neq 0$, with correlations decaying geometrically at the correlation length $\tau = -1/\ln |\alpha|$. The coupling is global although the mechanism is local; the locality is in the inverse.

Every frame is corrupted independently by the Ornstein–Uhlenbeck channel of Section 2.2, $dX_t = -X_t dt + \sqrt{\Delta_t} dW_t$ from $X_0 = a$, whose solution is $X_t | a \sim \mathcal{N}(ae^{-t}, \Delta_t)$ with $\Delta_t = 1 - e^{-2t}$ (Uhlenbeck and Ornstein, 1930; Gardiner, 2009). Stacking the L frames,

$$x = e^{-t}a + \sqrt{\Delta_t} \xi, \quad \xi \sim \mathcal{N}(0, I_L), \quad \xi \perp a, \quad (3.4)$$

and the object of study is the joint score of the noisy sequence,

$$S(x, t) = \nabla_x \log P_t(x) \in \mathbb{R}^L, \quad P_t(x) = \int_{\mathbb{R}^L} P_0(a) \prod_{k=0}^{L-1} \mathcal{N}(x_k; e^{-t}a_k, \Delta_t) da, \quad (3.5)$$

the gradient taken with respect to all L noisy frames at once (Figure 3.1).

Why the joint score. The smallest example already contains the argument. Two consecutive states of (3.2) have $\Sigma_0 = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}$; noising both coordinates independently through (3.4) leaves $\text{Var}(x_i) = e^{-2t} + \Delta_t = 1$ and gives $\text{Cov}(x_0, x_1) = \alpha e^{-2t} =: r_t$, so

$$\begin{pmatrix} x_0 \\ x_1 \end{pmatrix} \sim \mathcal{N}(0, \Sigma_t), \quad \Sigma_t = \begin{pmatrix} 1 & r_t \\ r_t & 1 \end{pmatrix}, \quad x_0, x_1 \sim \mathcal{N}(0, 1), \quad s_{t,k}^{\text{marg}}(x_k) = -x_k. \quad (3.6)$$

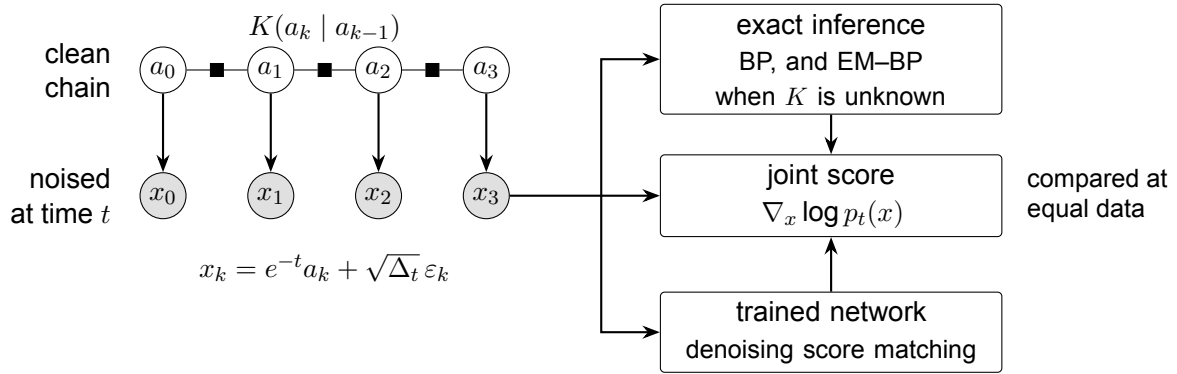


Figure 3.1: The object of the thesis and the two ways it is obtained. A Markov chain generates a clean trajectory; the OU channel corrupts every coordinate independently, so each observation attaches to the latent chain on its own and the posterior stays a chain. That makes the joint score exactly computable by message passing, with the kernel known (Chapters 4 and 5) or estimated from noised sequences (Chapter 6). The same score is what a denoising-score-matching network approximates without being given the chain; Chapter 8 measures what supplying it is worth.

Each one-frame marginal score contains no α at all. The joint score, from $\Sigma_t^{-1} = (1 - r_t^2)^{-1} \begin{pmatrix} 1 & -r_t \\ -r_t & 1 \end{pmatrix}$, is

$$s_t(x_0, x_1) = -\frac{1}{1 - r_t^2} \begin{pmatrix} x_0 - r_t x_1 \\ x_1 - r_t x_0 \end{pmatrix}. \quad (3.7)$$

The temporal coupling is invisible to every one-frame score, whose density carries zero Fisher information about α , and explicit in the joint score; by (2.8) computing the latter is an inference problem over the whole trajectory.

3.2 The clean law: the precision is tridiagonal

The score will depend not on Σ_0 but on its inverse, the *precision* $Q_0 := \Sigma_0^{-1}$.

Proposition 3.1 (Tridiagonal precision). *For the chain (3.2),*

$$Q_0 = \frac{1}{\sigma_\eta^2} \begin{pmatrix} 1 & -\alpha & & & 0 \\ -\alpha & 1 + \alpha^2 & -\alpha & & \\ & \ddots & \ddots & \ddots & \\ & & -\alpha & 1 + \alpha^2 & -\alpha \\ 0 & & & -\alpha & 1 \end{pmatrix}, \quad (Q_0)_{ij} = 0 \text{ for } |i - j| \geq 2. \quad (3.8)$$

Proof. For a centred Gaussian, $-2 \log P_0(a) = a^\top Q_0 a + \text{const}$, and the Markov factorisation (3.1) of (3.2) gives $-2 \log P_0(a) = a_0^2 + \sigma_\eta^{-2} \sum_{k=0}^{L-2} (a_{k+1} - \alpha a_k)^2 + \text{const}$. Reading the quadratic form off this sum: each product $a_k a_{k+1}$ appears once with coefficient $-2\alpha/\sigma_\eta^2$, no product of non-adjacent coordinates appears, an interior a_m^2 appears in two transition factors with total coefficient $(1 + \alpha^2)/\sigma_\eta^2$, and the endpoints appear in one, the left one together with its prior, $a_0^2(1 + \alpha^2/\sigma_\eta^2) = a_0^2/\sigma_\eta^2$ because $\sigma_\eta^2 = 1 - \alpha^2$. $\square$

The zeros have a direct meaning: for a jointly Gaussian vector, $(Q_0)_{ij} = 0$ if and only if $a_i \perp a_j$ given the remaining coordinates, so tridiagonality is the matrix expression of first-order Markov structure (for any smooth Markov density the Hessian of $-\log P$ vanishes at $|i - j| \geq 2$; here it is constant and equals Q_0). The same matrix predicts a hidden frame from the others, $\mathbb{E}[a_k \mid a_{\setminus k}] = -(Q_0)_{kk}^{-1} \sum_{l \neq k} (Q_0)_{kl} a_l$, which for an interior frame is

$$\mathbb{E}[a_k \mid a_{\setminus k}] = \frac{\alpha}{1 + \alpha^2} (a_{k-1} + a_{k+1}) : \quad (3.9)$$

all information the clean chain supplies about a_k is mediated by its two neighbours. This exact locality does not survive the channel at the level of the observed variables, which is the effect the rest of the chapter measures.

3.3 The noisy law, and the score two ways

Proposition 3.2 (Noised law). *Under (3.4) the noisy trajectory is Gaussian, $P_t(x) = \mathcal{N}(x; 0, \Sigma_t)$ with*

$$\boxed{\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I_L}, \quad (\Sigma_t)_{ij} = e^{-2t}\alpha^{|i-j|} + \Delta_t \delta_{ij}. \quad (3.10)$$

The proof is one line: x is a sum of independent Gaussians with covariances $e^{-2t}\Sigma_0$ and $\Delta_t I$. On the diagonal $e^{-2t} + \Delta_t = 1$, so every frame stays marginally $\mathcal{N}(0, 1)$ at every t (the channel is variance preserving); off the diagonal every entry is shrunk by the single factor e^{-2t} , so Σ_t is Toeplitz and dense at every finite t , interpolating from Σ_0 at $t = 0$ to I as $t \rightarrow \infty$. Since $\log P_t(x) = -\frac{1}{2}x^\top Q_t x + \text{const}$,

$$\boxed{S(x, t) = -Q_t x, \quad Q_t := \Sigma_t^{-1}.} \quad (3.11)$$

The joint score is linear in x , the signature of a Gaussian P_t , and its whole content is Q_t : row k says which frames the denoiser of frame k draws on and how strongly. At $L = 2$ this is (3.7). The important change is in the precision, not the covariance: at $t = 0$, $Q_t = Q_0$ is tridiagonal; for $t > 0$, adding $\Delta_t I$ to the covariance and inverting destroys that structure, and Q_t is dense.

The Bayesian route. The score–posterior identity (2.8) gives the same object as a posterior mean,

$$S_k(x, t) = \frac{e^{-t} \mathbb{E}[a_k | x] - x_k}{\Delta_t}, \quad (3.12)$$

and that posterior has a structure the marginal law lacks.

Proposition 3.3 (Posterior of the clean chain). $P(a | x) = \mathcal{N}(a; m, J^{-1})$ with

$$\boxed{J = Q_0 + \frac{e^{-2t}}{\Delta_t} I_L, \quad h = \frac{e^{-t}}{\Delta_t} x, \quad m = \mathbb{E}[a | x] = J^{-1}h.} \quad (3.13)$$

Proof. $-2 \log P(a | x) = a^\top Q_0 a + \Delta_t^{-1} \|x - e^{-t}a\|^2 + \text{const} = a^\top (Q_0 + e^{-2t} \Delta_t^{-1} I) a -$

$2(e^{-t}\Delta_t^{-1}x)^\top a + \text{const}$, which is a Gaussian in information form with precision J and field h . $\square$

The two routes agree exactly. From $J = \Delta_t^{-1}\Sigma_0^{-1}(\Delta_t I + e^{-2t}\Sigma_0) = \Delta_t^{-1}\Sigma_0^{-1}\Sigma_t$ one has $J^{-1} = \Delta_t Q_t \Sigma_0$, and substituting $m = e^{-t}\Delta_t^{-1}J^{-1}x$ into (3.12),

$$\frac{e^{-t}m - x}{\Delta_t} = \frac{1}{\Delta_t} \left[Q_t(e^{-2t}\Sigma_0) - I \right] x = \frac{1}{\Delta_t} \left[Q_t(\Sigma_t - \Delta_t I) - I \right] x = -Q_t x. \quad (3.14)$$

What (3.13) adds is structural: the channel enters J on the diagonal only, because conditional on a each observation depends on its own a_k alone, so the posterior precision of the clean chain stays tridiagonal at every t while the marginal precision Q_t of the noisy data is dense. The Bayesian route is the one that survives beyond the Gaussian case (Chapter 5), and the tridiagonal J is what makes it cheap (Chapter 4).

3.4 The spectral form

Between the tridiagonal Q_0 at $t = 0$ and the identity at $t = \infty$ lies the whole lifecycle of the couplings, and one tool resolves it: the forward map (3.10) is affine in Σ_0 . Write $\Sigma_0 = U \text{diag}(\omega_i)U^\top$ with U orthogonal and $\omega_i > 0$.

Theorem 3.4 (Spectral form and decoupling). *For every $t \geq 0$,*

$$\boxed{\Sigma_t = U \text{diag}(\lambda_i(t))U^\top, \quad Q_t = U \text{diag}(1/\lambda_i(t))U^\top, \quad \lambda_i(t) = e^{-2t}\omega_i + \Delta_t.} \quad (3.15)$$

In the coordinates $\tilde{x} = U^\top x$ the noisy law factorises, $P_t(\tilde{x}) = \prod_i \mathcal{N}(\tilde{x}_i; 0, \lambda_i(t))$, the score splits into scalar components $\tilde{S}_i = -\tilde{x}_i/\lambda_i(t)$, and the reverse-time dynamics decouple into L independent scalar linear Gaussian diffusions, one per mode.

Proof. $I = UU^\top$ is diagonal in every orthonormal basis, so $e^{-2t}\Sigma_0 + \Delta_t I = U \text{diag}(e^{-2t}\omega_i + \Delta_t)U^\top$, and every $\lambda_i(t) \geq \Delta_t > 0$ for $t > 0$ (and $\lambda_i(0) = \omega_i > 0$), so the inverse is diagonal in the same basis. Then $\text{Cov}(\tilde{x}) = U^\top \Sigma_t U$ is diagonal, an uncorrelated Gaussian is

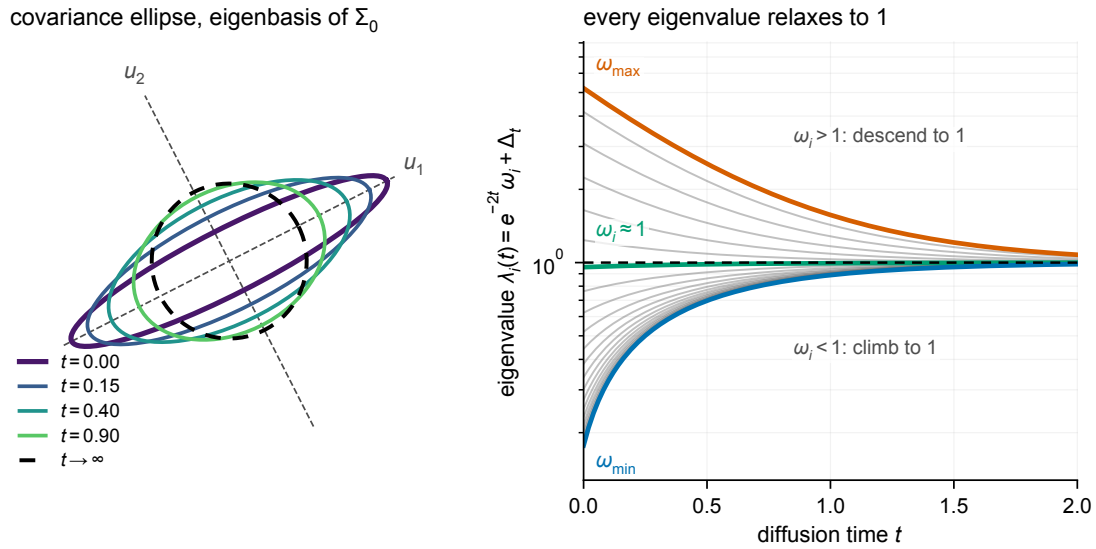


Figure 3.2: The spectral form (3.15) as geometry ($\alpha = 0.7$, $L = 24$). *Left:* the covariance ellipse in the eigenbasis of Σ_0 at four diffusion times; its axes are fixed and only the semi-axes $\sqrt{\lambda_i(t)}$ change until the ellipse is the unit circle. *Right:* the eigenvalue flow $\lambda_i(t)$ for all modes.

independent, $\tilde{S} = -U^T Q_t U \tilde{x}$, and an orthogonal image of white noise is white, so the reverse SDE (2.5) separates coordinate by coordinate. $\square$

Diffusion never rotates the problem: the eigenvectors are frozen for all t and only the eigenvalues move, each towards 1 (Figure 3.2). In the eigenbasis the chain is L scalar denoising problems with signal-to-noise ratios $e^{-2t}\omega_i/\Delta_t$; everything non-trivial in the frame basis, the dense Q_t and the band that fills and empties, is this diagonal object seen through the fixed rotation U .

Two rearrangements of (3.10) are used later. Factoring Δ_t gives $Q_t = \Delta_t^{-1}(I + \gamma_t \Sigma_0)^{-1}$ with the signal-to-noise ratio $\gamma_t = e^{-2t}/\Delta_t$ running from ∞ to 0, so $Q_t \approx e^{2t}Q_0$ for small t and $Q_t \approx I$ for large t : the precision is local at both ends of the schedule. Factoring e^{-2t} instead, with $c_t := e^{2t} - 1 \sim 2t$, gives the *resolvent form*

$$Q_t = e^{2t} Q_0 (I + c_t Q_0)^{-1} = e^{2t} \sum_{m=0}^{\infty} (-c_t)^m Q_0^{m+1} \quad (c_t \lambda_{\max}(Q_0) < 1), \quad (3.16)$$

the Neumann series converging near $t = 0$. Since Q_0 is tridiagonal, Q_0^m has bandwidth

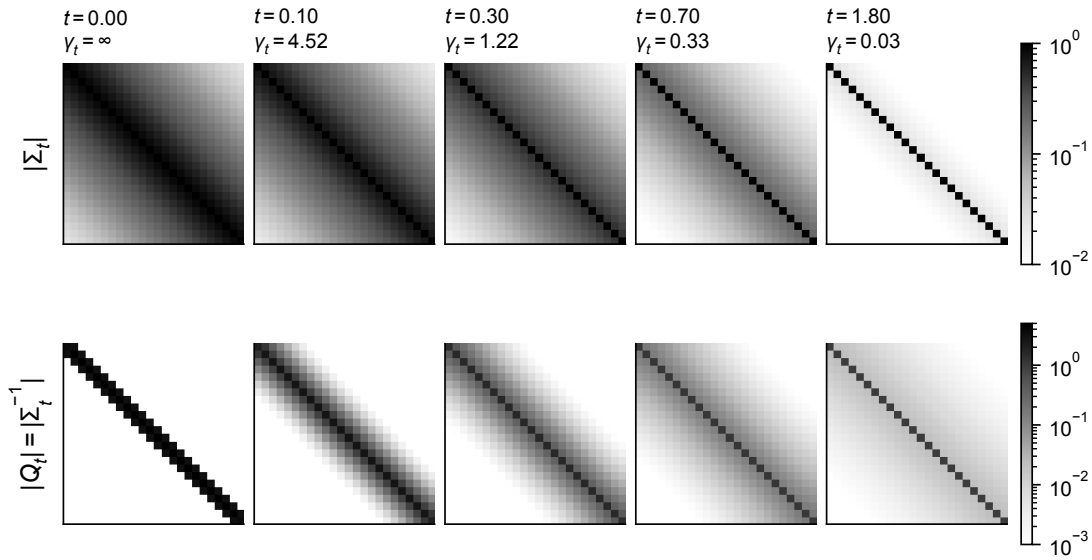


Figure 3.3: Magnitude heatmaps (log grey scale) of $|\Sigma_t|$ (top) and $|Q_t|$ (bottom), $\alpha = 0.85$, $L = 24$, at five diffusion times with their signal-to-noise ratios γ_t . At $t = 0$ the covariance is dense and the precision tridiagonal; as t grows the covariance contracts onto its diagonal while the precision band fills outward, then fades, until both are within $O(e^{-2t})$ of the identity.

m , so two frames at distance d are first coupled by the term Q_0^d , which enters with coefficient c_t^{d-1} : the dense precision does not appear everywhere at once, but fills outward from the diagonal one graph distance per order in t .

3.5 The lifecycle of the precision

Sparsity is a property of a matrix in a basis. In the eigenbasis the problem is diagonal at all times; in the frame basis, the operationally meaningful one, the zeros of Q_0 are a cancellation that diffusion destroys. The destruction answers three questions: when, how, and how much (Figure 3.3).

When: immediately. For $\alpha \neq 0$ every off-diagonal entry of Q_t is non-zero at every $t > 0$. Analyticity in t alone does not give this, since an analytic function can vanish at isolated times; the resolvent does. From (3.16), $Q_t = (e^{2t}/c_t)[I - (I + c_t Q_0)^{-1}]$, so for $i \neq j$ the entry $(Q_t)_{ij}$ is a multiple of an off-diagonal entry of the inverse of $B := I + c_t Q_0$, a symmetric positive-definite tridiagonal matrix with non-zero off-diagonal entries

$-c_t\alpha/\sigma_\eta^2$. The inverse of such an irreducible tridiagonal matrix has entries $(B^{-1})_{ij}$, $i \leq j$, equal to $(-1)^{i+j}$ times a product of the off-diagonal entries between i and j and of leading and trailing principal minors, all strictly positive because $B \succ 0$; no factor vanishes. The same formula gives the sign pattern: for $\alpha > 0$ all off-diagonal couplings of Q_t are negative (attractive at every distance), for $\alpha < 0$ they alternate with $|i - j|$.

How: one diagonal per order in t .

Theorem 3.5 (Band-fill leading order). *For small t and frame distance $d \geq 1$,*

$$(Q_t)_{i,i+d} = (-1)^{d-1} (2t)^{d-1} (Q_0^d)_{i,i+d} + O(t^d). \quad (3.17)$$

Proof. In (3.16) substitute $c_t = 2t + O(t^2)$ and $e^{2t} = 1 + O(t)$. The entry at distance d receives its first contribution from the single term $m = d - 1$, with prefactor $(-c_t)^{d-1} = (-1)^{d-1} (2t)^{d-1} (1 + O(t))$; every later term is $O(t^d)$. $\square$

How much: the off-tridiagonal mass. Structure is measured with the Frobenius norm $\|A\|_F = (\sum_{ij} A_{ij}^2)^{1/2}$, which is additive over disjoint entry sets. Let Π project onto the Markov band $\{|i - j| \leq 1\}$ that Q_0 occupies, so that ΠQ_t is the nearest-neighbour approximation of the score operator; then

$$\|Q_t\|_F^2 = \|\text{diag } Q_t\|_F^2 + \|\Pi Q_t - \text{diag } Q_t\|_F^2 + M_\perp(t)^2, \quad M_\perp(t) := \|Q_t - \Pi Q_t\|_F, \quad (3.18)$$

and $M_\perp$ is the norm of exactly the couplings a Markov model may not carry. At small t , (3.17) gives $(Q_t)_{i,i+2} = -2t(Q_0^2)_{i,i+2} + O(t^2)$, so $M_\perp$ leaves zero linearly in t . At large t , expanding $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$,

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}), \quad (3.19)$$

so every deviation from the identity decays at the single rate e^{-2t} and the score approaches the isotropic $-x$. In between, $M_\perp$ rises, peaks and falls (Figure 3.4): the

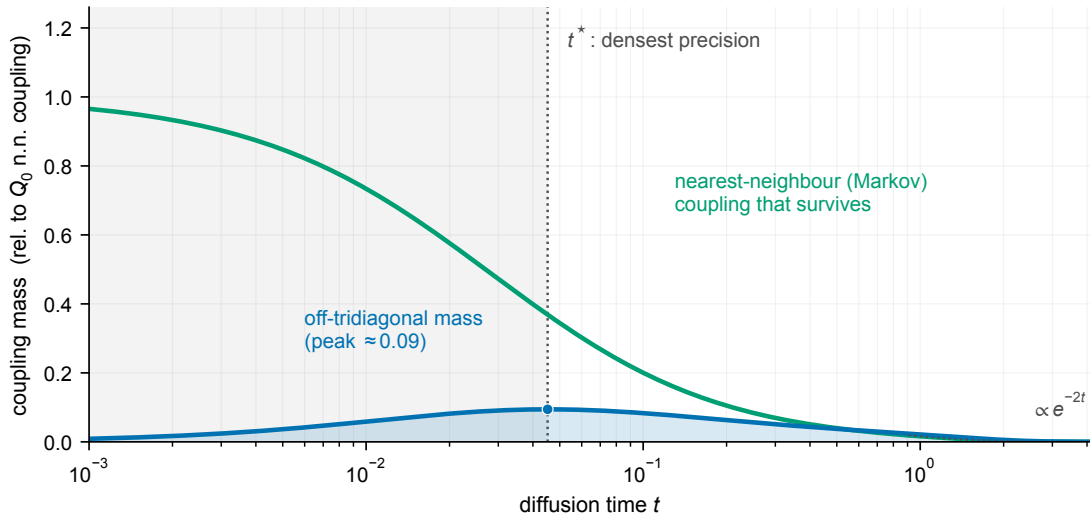


Figure 3.4: Frobenius lifecycle of the inter-frame coupling of Q_t ($\alpha = 0.9$, $L = 48$), each mass normalised by the nearest-neighbour coupling mass of Q_0 . The nearest-neighbour (Markov) mass decays monotonically, $\propto e^{-2t}$ once noise dominates. The off-tridiagonal mass rises from zero, peaks at $t^* \approx 0.05$, the least Markov point of the schedule, then decays with the rest of the structure; its peak is below 10% of the original coupling.

noisy sequence is most non-Markov at intermediate diffusion times, yet even at the peak the off-tridiagonal mass stays below one tenth of the clean nearest-neighbour mass, in this norm and at the displayed (α, L) . The lifecycle in one line: tridiagonal at $t = 0$; densifying at rate t^{d-1} per diagonal d ; maximally coupled at intermediate t ; isotropic as $t \rightarrow \infty$ at rate e^{-2t} . Chapter 4 asks what it costs to ignore the mass outside the band.

Chapter 4

The Gaussian Chain by Belief Propagation

Chapter 3 obtained the joint score by inverting the dense $L \times L$ matrix Σ_t , at $O(L^3)$. That cost belongs to the route, not to the problem: the posterior precision J of Proposition 3.3 is tridiagonal, so a structured solve delivers the same answer in $O(L)$. This chapter obtains the score by the message-passing form of that solve, sum-product on the factor graph of the posterior, generic and exact on any chain; for the Gaussian chain a message collapses to two numbers and the recursion is the Kalman filter followed by the Rauch-Tung-Striebel smoother. The chapter closes with the cost of locality: what truncating the evidence range does to the score.

4.1 Factor graphs and the sum-product algorithm

A factor graph makes a factorisation graphical (Kschischang et al., 2001): given $p(x) = Z^{-1} \prod_a \psi_a(x_{\partial a})$ it is the bipartite graph with a variable node per x_i , a factor node per ψ_a , and an edge whenever $x_i \in x_{\partial a}$. A connected factor graph with one fewer edge than nodes is a *tree*. Sum-product belief propagation (BP) passes messages along every edge (Pearl, 1988; Mézard and Montanari, 2009): variable-to-factor messages multiply the incoming factor messages from all *other* neighbours, and factor-to-variable

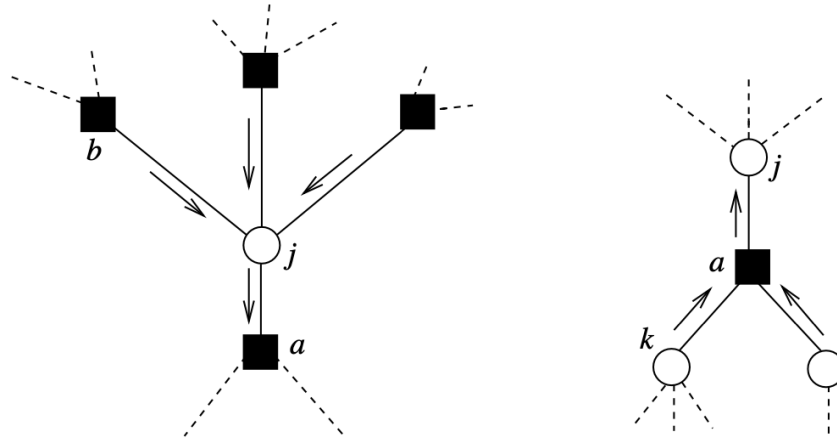


Figure 4.1: The two halves of (4.1). *Left:* the message from variable j to factor a , a function of the incoming factor messages from every other factor neighbour $b \neq a$. *Right:* the message from factor a to variable j , a function of the incoming variable messages from every other variable neighbour $k \neq j$. The original carries iteration superscripts; on a tree one inward and one outward sweep suffice. Reproduced from Mézard and Montanari (2009), their Figure 14.3.

messages integrate the factor against the incoming variable messages,

$$\nu_{i \rightarrow a}(x_i) \propto \prod_{b \in \partial i \setminus a} \hat{\nu}_{b \rightarrow i}(x_i), \quad \hat{\nu}_{a \rightarrow i}(x_i) \propto \int \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} \nu_{j \rightarrow a}(x_j) \, dx_{\partial a \setminus i}, \quad (4.1)$$

with the belief at node i proportional to $\prod_{a \in \partial i} \hat{\nu}_{a \rightarrow i}(x_i)$ (Figure 4.1). The “exclude the recipient” rule is the heart of the algorithm: the message $i \rightarrow a$ summarises everything node i knows except what it learned from a , so no evidence is counted twice along a path.

Theorem 4.1 (Tree exactness of sum–product). *On a tree-shaped factor graph, one inward and one outward pass of (4.1) compute the exact node and factor marginals of p (Mézard and Montanari, 2009, Theorem 14.1). The beliefs are unique; the unnormalised messages are defined up to positive constants.*

Proof sketch. Each edge of a tree splits the graph into two disjoint halves, so the message crossing it is exactly the partial sum over the half behind it, and no evidence reaches a node twice. Induction from the leaves inward terminates with every marginal

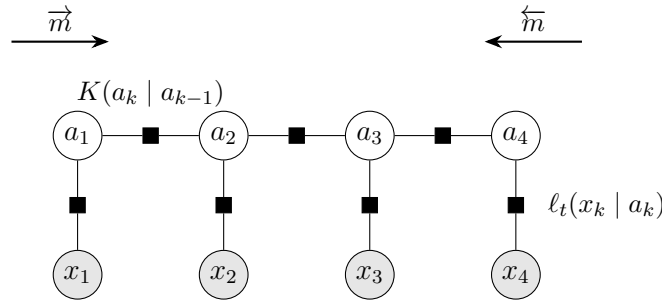


Figure 4.2: The posterior factor graph of a Markov chain observed through the coordinatewise channel. Open circles are latent variables, shaded circles observations, squares factors. Each transition factor joins two adjacent latent variables; each observation enters through a unary factor attached to a single latent variable. The posterior graph is therefore still a chain, Theorem 4.1 applies, and the arrows show the two message sweeps.

having absorbed each factor once. □

On graphs with cycles the same updates, iterated, define loopy BP, which is approximate (Weiss and Freeman, 2001); none of that is needed here. Bayes' rule multiplies the chain prior (3.1) by the per-frame likelihoods of (3.4), and nothing in the likelihood couples two frames, so for the Gaussian chain

$$p(a | x) \propto \underbrace{\psi_{\text{pr}}(a_0)}_{\mathcal{N}(a_0; 0, 1)} \prod_{k=0}^{L-2} \underbrace{\psi_{\text{tr}}^{(k)}(a_k, a_{k+1})}_{\mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2)} \prod_{k=0}^{L-1} \underbrace{\psi_{\text{ob}}^{(k)}(a_k)}_{\mathcal{N}(x_k; e^{-t} a_k, \Delta_t)}, \quad (4.2)$$

a hidden Markov model (Rabiner, 1989) with the data x_k absorbed into unary observation factors. Its factor graph has L variable nodes, $2L$ factor nodes and $3L - 1$ edges: a tree, a spine of variables linked by transition factors with one evidence leaf per node (Figure 4.2). Conditioning on the data did not change the graph of the prior, because independent per-node observation re-weights a node potential and never joins two spine nodes; correlated observation noise would add edges and break the construction. Nothing in this paragraph used Gaussianity: the posterior of *any* chain prior under coordinatewise noise is a tree, and Theorem 4.1 applies to it.

Tree exactness is a property of the graph and says nothing about whether the messages can be *represented*: for discrete variables a message is a finite vector, for continuous variables a density, and the recursion is made computable by closure in a finite-

dimensional family (Section 4.3), by a numerical grid (the reference solver of Chapter 5), or by projection onto a tractable family at a cost Chapter 5 measures.

4.2 The chain sweeps and their cost

On the tree of Figure 4.2 the schedule of (4.1) collapses to one forward and one backward pass.

Proposition 4.2 (Chain sweeps). *Define*

$$\mu_{\rightarrow 0}(a_0) := \psi_{\text{pr}}(a_0), \quad \mu_{\rightarrow k+1}(a_{k+1}) := \int \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\rightarrow k}(a_k) \, da_k, \quad (4.3)$$

$$\mu_{\leftarrow L-1}(a_{L-1}) := 1, \quad \mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\text{tr}}^{(k-1)}(a_{k-1}, a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) \, da_k. \quad (4.4)$$

Then for every k ,

$$p(a_k | x) \propto \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k), \quad (4.5)$$

where $\mu_{\rightarrow k}$ depends on $x_0, \dots, x_{k-1}$ only, $\mu_{\leftarrow k}$ on $x_{k+1}, \dots, x_{L-1}$ only, and the local evidence enters once.

Proof. Fix k and split the factors of (4.2) into those left of a_k , the local observation, and those right of a_k . With a_k held fixed the two sides share no variable, so integrating out everything else factorises into a left integral, which is $\mu_{\rightarrow k}(a_k)$ by definition, and a right integral, which is $\mu_{\leftarrow k}(a_k)$. Peeling off the last integration variable of the left integral gives (4.3); the first of the right integral gives (4.4). $\square$

For a chain of K -state variables every message is a vector in $\mathbb{R}^K$, one transition update is a $K \times K$ matrix–vector product, and a forward–backward sweep sends $O(L)$ of them: time $O(LK^2)$ and space $O(LK)$, linear in the sequence length.

4.3 The Gaussian closure: Kalman filtering and RTS smoothing

Two facts close the recursion algebraically: a product of Gaussians in the same variable is Gaussian, and an affine image of a Gaussian plus independent Gaussian noise is Gaussian. Write an unnormalised scalar Gaussian in natural form, $\mu(a) \propto \exp(-\frac{1}{2}\lambda a^2 + ra)$, with precision λ and field r , so mean r/λ and variance $1/\lambda$. The observation factor is itself Gaussian in a_k , $\psi_{\text{ob}}^{(k)}(a_k) \propto \exp(-\frac{1}{2}\lambda_{\text{ob}}a_k^2 + r_{\text{ob},k}a_k)$ with

$$\lambda_{\text{ob}} = \frac{e^{-2t}}{\Delta_t}, \quad r_{\text{ob},k} = \frac{e^{-t}x_k}{\Delta_t} : \quad (4.6)$$

a pseudo-observation at the deconvolved location $e^t x_k$ with precision e^{-2t}/Δ_t .

Theorem 4.3 (Gaussian closure of BP on the chain). *For the Gaussian AR(1) chain observed through the OU channel every sum-product message and every belief is Gaussian. With (λ, r) the natural parameters of the message arriving at site k and $(\bar{\lambda}, \bar{r}) = (\lambda + \lambda_{\text{ob}}, r + r_{\text{ob},k})$ after absorbing the local observation, the sweeps are*

$$\lambda_{\rightarrow k+1} = \frac{\bar{\lambda}}{\alpha^2 + \sigma_\eta^2 \bar{\lambda}}, \quad r_{\rightarrow k+1} = \frac{\alpha \bar{r}}{\alpha^2 + \sigma_\eta^2 \bar{\lambda}}; \quad \lambda_{\leftarrow k-1} = \frac{\alpha^2 \bar{\lambda}}{1 + \sigma_\eta^2 \bar{\lambda}}, \quad r_{\leftarrow k-1} = \frac{\alpha \bar{r}}{1 + \sigma_\eta^2 \bar{\lambda}}, \quad (4.7)$$

and the smoothing belief at site k adds the three natural parameters,

$$\lambda_k^{\text{post}} = \lambda_{\rightarrow k} + \lambda_{\text{ob}} + \lambda_{\leftarrow k}, \quad r_k^{\text{post}} = r_{\rightarrow k} + r_{\text{ob},k} + r_{\leftarrow k}, \quad \mathbb{E}[a_k | x] = \frac{r_k^{\text{post}}}{\lambda_k^{\text{post}}}, \quad (4.8)$$

from which (3.12) gives the score. Each update is $O(1)$, so all posterior marginals, and hence the complete joint score, are obtained exactly in $O(L)$ time and space.

Proof. Multiplying Gaussian factors adds natural parameters, which gives $(\bar{\lambda}, \bar{r})$. Forward, $a_k \sim \mathcal{N}(\bar{r}/\bar{\lambda}, 1/\bar{\lambda})$ pushed through $a_{k+1} = \alpha a_k + \eta_k$ is $\mathcal{N}(\alpha \bar{r}/\bar{\lambda}, \sigma_\eta^2 + \alpha^2/\bar{\lambda})$, whose natural parameters are the first pair in (4.7). Backward, $\int \mathcal{N}(a_k; \alpha a_{k-1}, \sigma_\eta^2) \exp(-\frac{1}{2}\bar{\lambda}a_k^2 + \bar{r}a_k) da_k$ is a Gaussian function of a_{k-1} with the second pair. The belief (4.5) is a product

of three Gaussians. No approximation is introduced at any step: the family is closed under the two operations the algorithm uses, and the graph is a tree. $\square$

Run in mean–variance arithmetic the forward sweep is the Kalman filter, predict $(m, v) \mapsto (\alpha m, \alpha^2 v + \sigma_\eta^2)$ followed by a precision-weighted update with the observation, and combining it with the backward sweep is the Rauch–Tung–Striebel smoother, update for update (Kalman, 1960; Rauch et al., 1965): the two classical literatures compute one object. The closure is not the Gaussian approximation of TAP or approximate message passing (Thouless et al., 1977; Donoho et al., 2009), which compress many weak incoming messages into Gaussian updates with an Onsager correction in a large-system limit; here the messages are Gaussian because multiplication and marginalisation are exact operations on the family, on a tree of finite degree. Tree exactness holds for any innovation law while closure holds only because the model is Gaussian, and Chapter 5 is about the second property failing while the first survives. Numerically, the audit suite runs BP against the matrix route (3.11) over 80 configurations ($L \in \{2, 3, 5, 8, 16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$): maximum disagreement 1.3×10^{-15} on posterior means and 1.2×10^{-14} on scores, double-precision rounding accumulated along the recursion, reproduced by an independent implementation written from the recursion alone (Appendix A). One qualitative detail from an $L = 3$ instance: at $x = (1.2, -0.4, 0.8)$ the score component S_1 is positive although $x_1 < 0$, because the neighbours pull the posterior mean of the middle frame positive, which no per-frame score can produce.

4.4 The price of locality

Belief propagation on the chain is exact but not local: the message reaching frame k has travelled the whole sequence, and Q_t is dense for every $t > 0$. How much of the score survives if evidence is allowed to travel only a few frames? There are two ways to go local.

Truncate the answer. Keep the entries of the exact operator within a band and use

what is left,

$$\widehat{S}^{(b)}(x, t) = -Q_t^{(b)}x, \quad (Q_t^{(b)})_{ij} = (Q_t)_{ij} \mathbf{1}\{|i - j| \leq b\}; \quad (4.9)$$

$b = 1$ is the tridiagonal, Markov, truncation of the score. *Truncate the algorithm.* Let the messages of Proposition 4.2 travel at most r hops before frame k is read off, replacing the message beyond the cut by the stationary prior marginal. Because a contiguous window of a stationary AR(1) chain is itself a stationary AR(1) chain, this is exact inference on the window,

$$m_k^{(r)} = \mathbb{E}[a_k \mid x_{k-r}, \dots, x_{k+r}], \quad \widehat{S}_k^{(r)}(x, t) = \frac{e^{-t}m_k^{(r)} - x_k}{\Delta_t}. \quad (4.10)$$

Both estimators are linear, $\widehat{S} = -Mx$, so their error against $S = -Q_t x$ can be averaged over the data law $x \sim \mathcal{N}(0, \Sigma_t)$ without sampling. Using $\mathbb{E}[x^\top A x] = \text{tr}(A \Sigma_t)$ and $\text{tr}(Q_t \Sigma_t Q_t) = \text{tr} Q_t$, the relative RMS score error is the closed form

$$\varepsilon(M) := \sqrt{\frac{\mathbb{E}\|(M - Q_t)x\|^2}{\mathbb{E}\|Q_t x\|^2}} = \sqrt{\frac{\text{tr}[(M - Q_t) \Sigma_t (M - Q_t)^\top]}{\text{tr} Q_t}}, \quad (4.11)$$

a population error, not a sampled estimate: $\varepsilon = 0.1$ means the RMS approximation error is a tenth of the RMS size of the exact score. Table 4.1 and Figure 4.3 evaluate it for both families.

Table 4.1: Relative RMS score error (4.11) of the two truncations ($L = 40$, $\alpha = 0.8$); every entry is closed-form. The last two rows are derived: the fraction of the no-neighbour error removed by admitting a single hop, and the penalty for truncating the answer rather than the algorithm at equal budget.

relative RMS score error	$t = 0.05$	$t = 0.3$	$t = 1$	$t = 3$
no neighbours, $r = 0$	0.815	0.567	0.194	0.0045
one hop, $r = 1$	0.123	0.235	0.130	0.0035
two hops, $r = 2$	0.020	0.107	0.089	0.0028
banded answer, $b = 1$	0.210	0.302	0.135	0.0035
banded answer, $b = 2$	0.032	0.134	0.092	0.0028
removed by the first hop	85%	59%	33%	21%
penalty, $b=1$ over $r=1$	$1.71\times$	$1.28\times$	$1.03\times$	$1.00\times$

The first row is the cost of treating the frames as independent: at $t = 0.05$ the isolated-frame score is wrong by 82% in relative RMS, so almost all of the score at small diffusion times *is* inter-frame structure, and the error does not vanish as $t \rightarrow 0$ because the $r = 0$ estimator converges to $-x$ while the exact score converges to $-Q_0x$. A single nearest-neighbour exchange, the Markov band and nothing more, removes 85% of it, and two hops remove 98%. The removed fraction falls with diffusion time, but the first row shows why the band is not mattering less: by $t = 3$ the entire inter-frame content of the score is worth 0.0045 in relative error, so nothing is left for any hop to recover. The band carries the largest share of the coupling exactly where the coupling is large. At equal budget, truncating the inference beats truncating the score matrix at small t (0.123 against 0.210 at $t = 0.05$), and the gap closes as t grows. The error of the radius- r estimator decays geometrically in r at a rate that worsens with t ; the closed-form influence-decay rate behind that geometry is derived in Chapter 8, where it predicts the best receptive field of a learned score head.

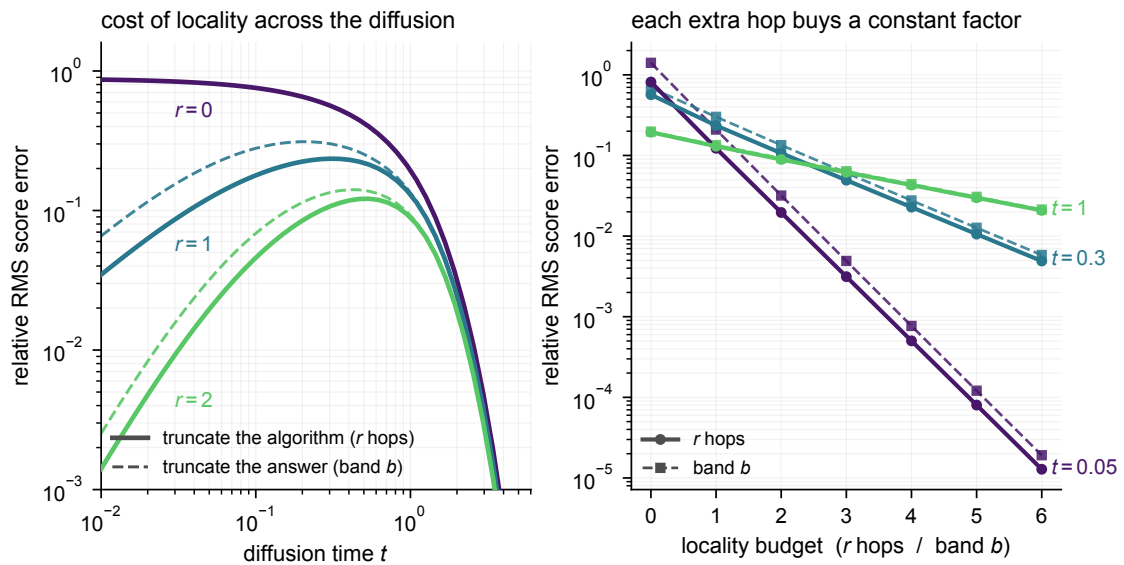


Figure 4.3: The two truncations ($L = 40$, $\alpha = 0.8$; every value closed-form by (4.11)). *Left:* relative RMS score error against diffusion time, for message range r (solid) and matrix band b (dashed). For $r \geq 1$ both are worst at intermediate t and vanish at both ends, at $t \rightarrow 0$ because the clean score is tridiagonal; the no-neighbour estimator is the exception, since the clean score of frame k genuinely uses its neighbours whenever $\alpha \neq 0$. *Right:* the same error against the locality budget at three diffusion times: geometric decay at a rate that worsens with t , with the banded answer (squares) above the windowed inference (circles) at every budget.

Chapter 5

Beyond Gaussianity: Laplace Innovations and the Gaussian-Message Approximation

Every exact result of Chapters 3 and 4 rests on the closure of the Gaussian family under the two operations the model performs, multiplying densities and marginalising. This chapter removes that property in the most controlled way available and answers RQ3: the AR(1) recursion, the OU channel and the factor-graph structure are kept, and only the innovation law changes, from Gaussian to Laplace.

5.1 What changes, and what is preserved

The Laplace chain keeps the initial law and the second-order structure of (3.2) fixed,

$$a_0 \sim \mathcal{N}(0, 1), \quad a_k = \alpha a_{k-1} + \varepsilon_k, \quad \varepsilon_k \sim \text{Laplace}(0, b) \text{ i.i.d.}, \quad b = \sqrt{(1 - \alpha^2)/2}, \quad (5.1)$$

so that $\text{Var}(\varepsilon_k) = 1 - \alpha^2$, the chain has unit marginal variance and $\text{Cov}(a_j, a_k) = \alpha^{|j-k|}$ exactly as before. The Gaussian and Laplace chains agree on every second moment and differ only beyond them. The score–posterior identity holds for any prior, so with $\mu := e^{-t}$,

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k | x] - x_k}{\Delta_t}; \quad (5.2)$$

the posterior still factorises over the tree of Figure 4.2, so the sweeps (4.3)–(4.4) still compute exact smoothing marginals, and the OU likelihood factors are still Gaussian in a_k . What is lost is that the messages are no longer Gaussian and the posterior mean, hence the score, is no longer linear in x (Figure 5.1).

The two chains first differ at fourth order. A centred Laplace variable of variance v has fourth cumulant $3v^2$, excess kurtosis 3; the chain *state*, a weighted sum of one Gaussian seed and k Laplace innovations, inherits a milder and position-dependent value, since fourth cumulants of independent sums add with the fourth power of the coefficients:

$$\kappa_4(a_k) = 3(1 - \alpha^2)^2 \sum_{m=0}^{k-1} \alpha^{4m} = 3 \frac{1 - \alpha^2}{1 + \alpha^2} (1 - \alpha^{4k}), \quad \boxed{\kappa_4(x_{t,k}) = e^{-4t} \kappa_4(a_k)} \quad (5.3)$$

the second identity because the channel adds an independent Gaussian with $\kappa_4 = 0$. At $\alpha = 0.85$ the bulk value is ≈ 0.48 , not 3; the chain is covariance-stationary by construction but not strictly stationary, because the Gaussian initial law is not the stationary law of the Laplace recursion; and Laplace-driven non-Gaussianity survives every finite diffusion time but decays as e^{-4t} , while the second moment is preserved exactly. Both facts return in Chapters 7 and 8.

5.2 Where closed forms stop

Theorem 5.1 (Joint score from chain messages). *For any chain prior, in particular (5.1), and the OU likelihood, the joint score is (5.2) with*

$$\mathbb{E}[a_k \mid x] = \frac{\int a_k \vec{m}_k(a_k) \psi_{\text{ob}}^{(k)}(a_k) \overleftarrow{m}_k(a_k) \mathrm{d}a_k}{\int \vec{m}_k(a_k) \psi_{\text{ob}}^{(k)}(a_k) \overleftarrow{m}_k(a_k) \mathrm{d}a_k}, \quad (5.4)$$

the messages being those of Proposition 4.2.

Computational locality survives; what does not survive is the collapse of a message

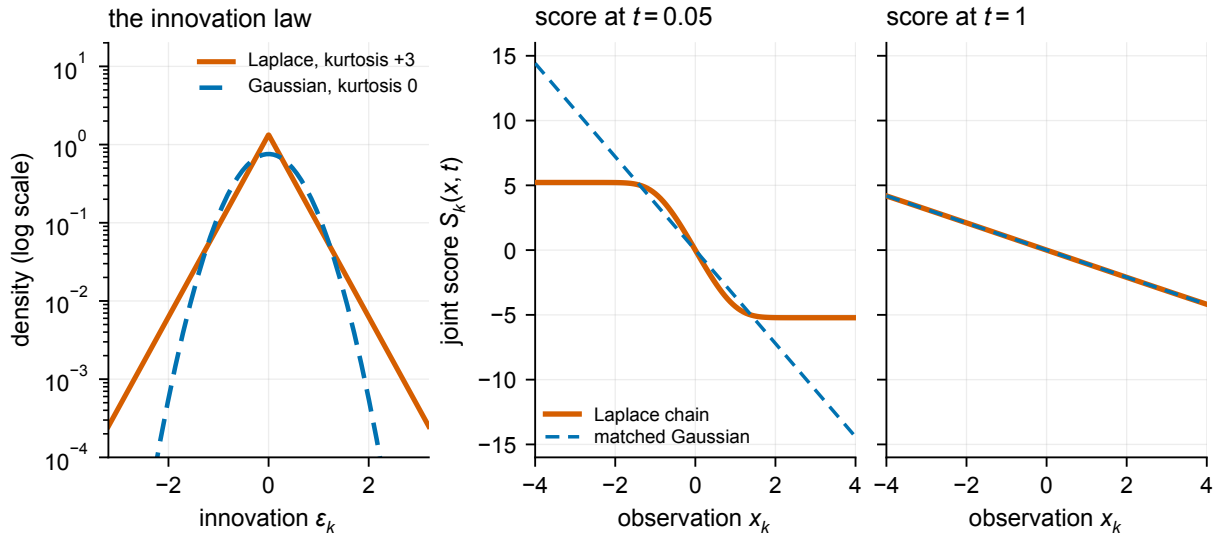


Figure 5.1: Laplace (vermilion, solid) against Gaussian (blue, dashed) at $\alpha = 0.85$. *Left:* the two innovation laws on a log scale, same variance $1 - \alpha^2$, different shape. *Centre and right:* the joint score (5.2) at the middle frame of a chain of $L = 9$ as x_k is swept with the other frames at zero, by grid BP (Section 5.3) against the covariance-matched Gaussian chain, which is exactly what the Gaussian closure returns (Theorem 5.2). At $t = 0.05$ the Laplace score saturates near ± 5.1 while the Gaussian one grows without bound; at $t = 1$ the two coincide on the same scale.

to two parameters. The first message can still be written down: with $M(a' | a) = (2b)^{-1} e^{-|a' - \alpha a|/b}$ and $y = x' - \mu \alpha a$, splitting the convolution $\int M(a' | a) \mathcal{N}(x'; \mu a', \Delta_t) da'$ at the single kink and completing the square on each half-line gives

$$I(y) = \frac{e^{\Delta_t/(2\mu^2 b^2)}}{2\mu b} \left[e^{-y/(\mu b)} \Phi\left(\frac{y}{\sqrt{\Delta_t}} - \frac{\sqrt{\Delta_t}}{\mu b}\right) + e^{y/(\mu b)} \Phi\left(-\frac{y}{\sqrt{\Delta_t}} - \frac{\sqrt{\Delta_t}}{\mu b}\right) \right], \quad (5.5)$$

with Φ the standard Gaussian cdf. Closure is lost at the very next update: the incoming message is now a sum of terms $e^{ca} \Phi(da + e)$, and integrating it against the next Laplace transition requires half-line integrals of $e^{-qa^2 + \ell a} \Phi(da + e)$, higher-dimensional Gaussian probabilities outside the normal–Laplace family, so each step increases the functional complexity; changing to innovation coordinates only moves the dependence from the prior into a dense likelihood. Exact BP remains valid, and its messages must be represented numerically.

5.3 Grid belief propagation

A message is a density on $\mathbb{R}$. It is represented by its values on N_g equispaced points $g_i = -A + ih$, $h = 2A/(N_g - 1)$, of a truncated interval $[-A, A]$, and every integral of (4.3)–(4.4) becomes a composite trapezoidal quadrature with weights $w_i = h$ in the interior and $h/2$ at the two endpoints. The transition enters as a matrix and the likelihood as a diagonal, $K_{ij} = M(g_i | g_j)$ and $(\ell_k)_i = \mathcal{N}(x_k; \mu g_i, \Delta_t)$, and with $\odot$ the elementwise product the sweeps read

$$f_0 \propto p_0 \odot \ell_0, \quad f_k \propto \ell_k \odot K(w \odot f_{k-1}), \quad k = 1, \dots, L-1, \quad (5.6)$$

$$\beta_{L-1} = \mathbf{1}, \quad \beta_k \propto K^\top(w \odot \ell_{k+1} \odot \beta_{k+1}), \quad k = L-2, \dots, 0, \quad (5.7)$$

with belief $\text{bel}_k = f_k \odot \beta_k$, posterior mean $\hat{m}_k = \sum_i w_i g_i (\text{bel}_k)_i / \sum_i w_i (\text{bel}_k)_i$ and score $\hat{S}_k = (\mu \hat{m}_k - x_k) / \Delta_t$. Each step is one matrix–vector product, so the solver costs $O(LN_g^2)$ time and $O(LN_g)$ space: sum–product over a finite alphabet, the alphabet now a discretisation. Three choices fix the representation (Table 5.1).

Table 5.1: The three choices that fix grid BP, what each is set to, and what fixes it.

choice	setting	what fixes it
renormalisation	forward messages to unit mass, backward to unit maximum, after every step	free, since messages are defined up to a constant (Theorem 4.1) and the ratio $\hat{m}_k$ is invariant; necessary, since without it a chain of $L = 32$ underflows
truncation A	$A = 8$	the exponential tail beyond $ a = 8$ carries below 10^{-9} of the mass for every α used; the mass in the outermost cells is logged per run
resolution N_g	$N_g = 401$ (reference 801)	the likelihood in a -coordinates has width $\sqrt{\Delta_t}/\mu \approx \sqrt{2t}$ at small t ; the gate
		$\sqrt{2t} \gtrsim 3h = 6A/(N_g - 1) \quad (5.8)$ keeps it wider than the spacing, and it is this gate, not the innovation law, that sets the smallest t reported

Validation. Grid BP is a numerical approximation of the functional recursion and serves as a reference only inside a validated regime, established on the Gaussian chain where Chapter 3 gives the truth. For smooth, rapidly decaying integrands the trapezoidal rule converges superalgebraically, so failure is abrupt, and the measurement shows exactly that: at $N_g = 101$, $A = 8$ the relative score error sits at the double-precision floor down to $t \approx 0.03$, is 10^{-10} at $t = 0.02$ and reaches the percent scale by $t = 0.005$. The gate (5.8) is conservative relative to that onset, and the working $N_g = 401$ clears it with room to spare ($3h = 0.12$ against $\sqrt{2t} = 0.32$ at the smallest reported $t = 0.05$). For the Laplace kernel the integrand has a kink and no spectral statement is invoked; the reference is certified by self-convergence, the error between $N_g = 401$ and 801 staying 10^4 to 10^7 times smaller than every effect measured below (Figure 5.3, left). Self-convergence cannot test the domain width, which the half-width sweep of `exp_18` (144 cells over

$A \in \{4, 6, 8, 10\}$) settles: the posterior mass reaching the outermost cells is 4.4×10^{-3} at $A = 4$, 4.2×10^{-4} at $A = 6$, 3.9×10^{-6} at the working $A = 8$ and 3.2×10^{-8} at $A = 10$, two orders below the smallest closure error reported here. Within this regime grid BP is treated as the reference score; outside it no claim is made.

One identity governs how everything is reported. For any two solvers with posterior means $\hat{m}$ and m_{ref} , (5.2) gives exactly

$$\hat{s} - s_{\text{ref}} = \frac{e^{-t}}{\Delta_t} (\hat{m} - m_{\text{ref}}), \quad \frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t} \quad (t \downarrow 0) : \quad (5.9)$$

every score error is an amplified posterior-mean error, its residual logged in every experiment between 10^{-15} and 10^{-12} , and small diffusion times are the difficult regime because whatever an approximation does to the posterior mean is amplified by $1/2t$ in the score.

5.4 The Gaussian-message approximation, measured

Gaussian-projected BP runs the same forward–backward recursion as grid BP but projects every transition message onto the Gaussian family by moment matching: $\Pi_G f$ is the Gaussian with the mean and variance of $f/\int f$. The first result says what that computes.

Theorem 5.2 (Gaussian closure is the LMMSE estimator). *Consider moment-projected sum–product on the chain (5.1) with $\alpha \neq 0$, Gaussian unary likelihoods, $\mathbb{E}[a_0] = 0$, innovations of mean zero and variance σ_η^2 , and a Gaussian (or projected) initial message. Its posterior means coincide exactly with those of the covariance-matched Gaussian model,*

$$\hat{a}_{\text{LMMSE}} = \text{Cov}(a, x) \text{Cov}(x)^{-1} x = e^{-t} \Sigma_0 (e^{-2t} \Sigma_0 + \Delta_t I)^{-1} x. \quad (5.10)$$

Hence the closure returns the same estimator for every innovation law sharing the moments $(0, \sigma_\eta^2)$.

Proof. By induction every projected message equals the corresponding exact message

of the covariance-matched Gaussian chain. The initial messages coincide by hypothesis. Multiplication by the Gaussian likelihood is the same precision-weighted update in both models. A Gaussian message with moments (m, v) pushed through $a_i = \alpha a_{i-1} + \varepsilon_i$ has moments $(\alpha m, \alpha^2 v + \sigma_\eta^2)$, and the backward transition integral, viewed as a density in the preceding state, has moments $(m/\alpha, (v + \sigma_\eta^2)/\alpha^2)$; both maps depend on the innovation law only through $(0, \sigma_\eta^2)$, so after Π_G the projected message is exactly the Gaussian chain's. (The terminal message $\overleftarrow{m}_{L-1} \equiv 1$ is not normalisable; the backward induction starts at the terminal likelihood, the first object actually propagated, and $\alpha \neq 0$ is used in the backward map.) All beliefs therefore coincide with the Gaussian chain's, whose posterior mean is $\text{Cov}(a, x) \text{Cov}(x)^{-1} x$ with $\text{Cov}(a, x) = e^{-t} \Sigma_0$ and $\text{Cov}(x) = e^{-2t} \Sigma_0 + \Delta_t I$. $\square$

Gaussian-projected BP is therefore not a heuristic Gaussianisation but the best second-order linear estimator, and comparing it with grid BP measures the loss from discarding structure beyond the first two moments. The *estimator* is the same for every innovation law with matched moments; its *error* against the true posterior is not, because each law has its own nonlinear posterior, and Section 5.4 measures how differently the same estimator misses different truths.

Three metrics. The relative score error is what the diffusion model consumes, but by (5.9) it is the posterior-mean error multiplied by e^{-t}/Δ_t and divided by the norm of the reference score, both of which move by orders of magnitude across the schedule, so on its own it cannot distinguish an approximation that improves from one whose errors are scaled away. The primary metric is therefore the posterior mean, which carries a risk interpretation.

Proposition 5.3 (Excess-risk decomposition). *For $m(x) = \mathbb{E}[a \mid x]$ and any estimator $\hat{a}(x)$ with finite second moment, $\mathbb{E}\|a - \hat{a}(x)\|^2 = \mathbb{E}\|a - m(x)\|^2 + \mathbb{E}\|m(x) - \hat{a}(x)\|^2$.*

Proof. Expand $a - \hat{a} = (a - m) + (m - \hat{a})$; the cross term vanishes after conditioning on x , since $\mathbb{E}[a - m(x) \mid x] = 0$ and both m and $\hat{a}$ are functions of x . $\square$

With $\hat{a} = \hat{a}_{\text{LMMSE}}$ the gap between the two posterior means is exactly the excess denoising risk of using second moments only. It is reported normalised by the signal energy $\mathbb{E}\|a\|^2 = L$,

$$\varepsilon_{\text{PM}} = \sqrt{\frac{1}{L} \mathbb{E}\|\hat{a}_{\text{LMMSE}} - \mathbb{E}[a | x]\|^2}, \quad (5.11)$$

dimensionless, free of the prefactor and stable at both ends of the schedule. Two secondary metrics accompany it: the excess-risk fraction $\mathbb{E}\|m - \hat{a}\|^2 / \mathbb{E}\|a - m\|^2$, the percentage by which closure raises the denoising error above the floor set by the posterior itself, and the relative score error, because it is what the generative process sees.

Table 5.2: Gaussian closure against validated grid BP on the Laplace chain ($\alpha = 0.85$, $L = 32$, 60 trials per t , reference $N_g = 801$), one sweep sampled densely for $t \lesssim 0.3$. Left group: the posterior approximation before the prefactor, ε_{PM} (5.11) with its 10–90% band, the same normalised to $t = 0.02$, and the excess MSE of Proposition 5.3 as a percentage of the MMSE floor. Last column: the resulting discrepancy in the score.

t	posterior inference					score
	ε_{PM}	10–90%	$\varepsilon_{\text{PM}}/\varepsilon_{\text{PM}}^{0.02}$	excess MSE [% floor]	MMSE floor	rel. score error
0.02	0.036	(0.029, 0.052)	1.00	3.9	0.032	0.38
0.032	0.040	(0.034, 0.057)	1.14	3.4	0.047	0.30
0.05	0.050	(0.036, 0.079)	1.42	3.9	0.066	0.27
0.072	0.053	(0.038, 0.082)	1.50	3.1	0.089	0.20
0.105	0.050	(0.037, 0.100)	1.41	2.1	0.114	0.14
0.153	0.043	(0.034, 0.085)	1.20	1.2	0.151	0.10
0.222	0.038	(0.033, 0.058)	1.06	0.74	0.193	0.065
0.322	0.033	(0.027, 0.059)	0.93	0.43	0.253	0.041
0.98	0.016	(0.012, 0.028)	0.45	0.045	0.584	7.3×10^{-3}
2.07	0.0018	(0.0013, 0.0038)	0.05	< 0.01	0.916	2.4×10^{-4}

The measurement. Table 5.2 and Figures 5.2 and 5.3 report the sweep, and two readings of it disagree. Read the last column and the closure improves steadily as noise

is added: the relative score error falls by a factor of 3.6 between $t = 0.02$ and $t = 0.15$. Read ε_{PM} and it does not: the posterior-mean error *rises* by half over $t \in [0.02, 0.07]$, is still larger at $t = 0.15$ than at $t = 0.02$, and begins to decay only past $t \simeq 0.3$. The gap between the readings is the prefactor of (5.9) and the shrinking norm of the reference score, neither of which reports on the inference. As a denoiser, the excess over the MMSE floor falls from 3.9% at $t = 0.05$ to 0.43% at $t = 0.32$ and below 0.05% once $t \gtrsim 1$, while the floor itself grows from 0.07 to 0.92: the approximation error shrinks as the irreducible uncertainty grows. Direction survives magnitude: the probability that a trial exceeds 10% relative score error is 1.00 through $t \simeq 0.07$ and zero by $t \simeq 0.7$, whereas the median cosine similarity to the reference score is already 0.964 at $t = 0.05$ and above 0.99 by $t \simeq 0.11$. Gaussian closure loses how strongly to move before it loses where to move. The peak in ε_{PM} separates two easy regimes: as $t \rightarrow 0$ each latent state is pinned by its own near-noiseless observation, and at large t the Gaussian noise washes out the higher-order structure of the innovation law; the largest error sits where the evidence is weak enough for the prior's shape to matter and strong enough not to have been erased. The Laplace posterior is log-concave and unimodal at every $t > 0$, so its worst trials are heavy-tailed and kinked rather than bimodal; this is a qualitative mechanism for the aggregate curve, not a measurement of posterior non-Gaussianity.

Varying the innovation law. Holding $\text{Cov}(a_i, a_j) = \alpha^{|i-j|}$ fixed, the innovation law is varied over Laplace, Student- t with $\nu \in \{5, 8\}$, and symmetric Gaussian mixtures of increasing separation. Across every family substantial closure error is confined to $t \lesssim 0.4$ and falls to about 10^{-3} or below by $t = 2.4$. At $t = 0.05$ and $\alpha = 0.5$ the median relative score error is 0.94 for a strongly bimodal mixture, 0.52 for Laplace, 0.32 for Student- t with $\nu = 5$, and 0.07 for a mildly bimodal mixture. Strong bimodality is the hardest case, a single Gaussian sitting between two separated posterior modes. Stronger chain correlation helps: for Laplace innovations at $t = 0.05$ the median error falls from roughly one half at $\alpha = 0.5$ to 0.27 at $\alpha = 0.85$ and 0.13 at $\alpha = 0.95$, so informative neighbours reduce the weight the local non-Gaussian shape carries, an

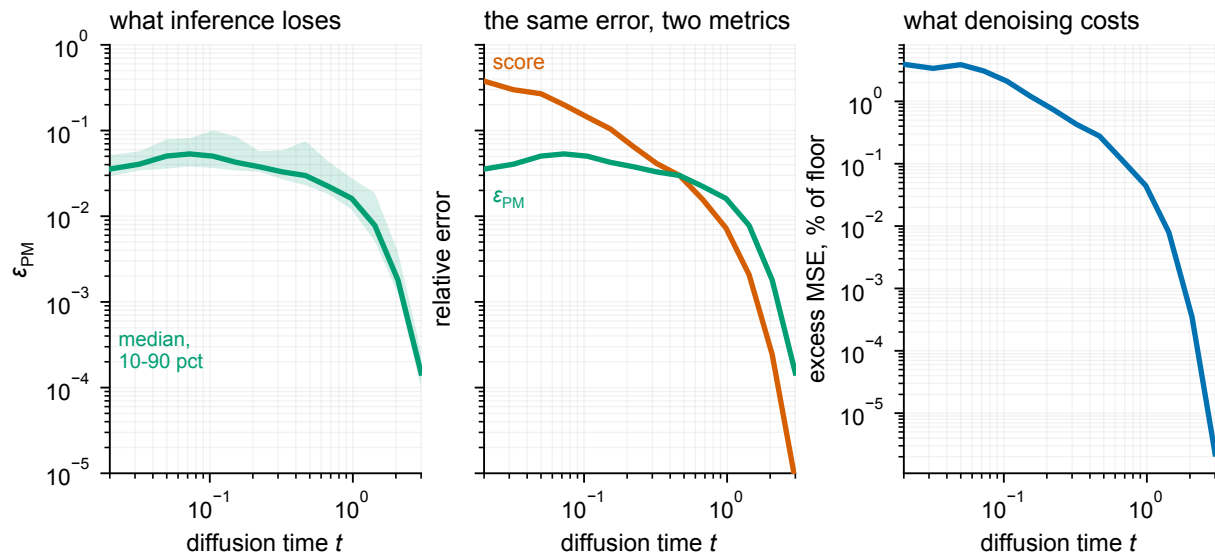


Figure 5.2: The same experiment under three metrics ($\alpha = 0.85$, $L = 32$, 60 trials per t). *Left:* the posterior-mean error (5.11), median and 10–90% band, not monotone: it peaks at $t \simeq 0.07$. *Centre:* the same runs through the relative score error, which falls by $3.6\times$ over the interval on which ε_{PM} is flat. *Right:* the excess denoising MSE of Proposition 5.3 as a percentage of the MMSE floor.

interpretation rather than a measured mechanism. The estimator is fixed by the model class; the size of its error is each law's own.

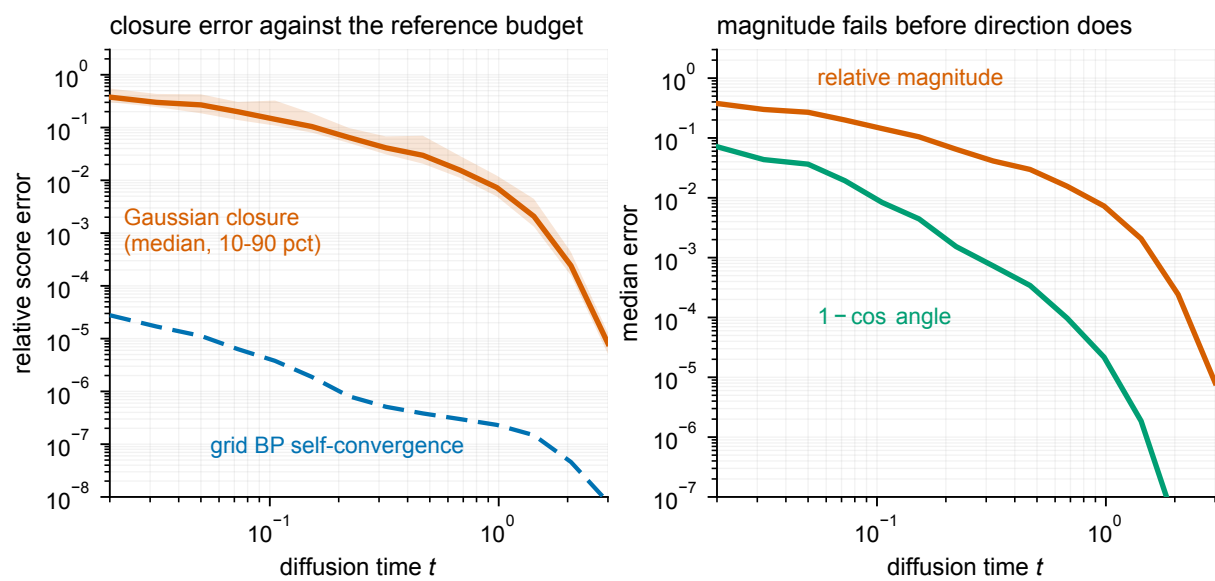


Figure 5.3: The score-side view. *Left:* median relative score error with its 10–90% band, against the grid-BP self-convergence error ($N_g = 401$ versus 801) on identical data; the closure error exceeds the numerical budget by three to five orders of magnitude, so the measurement is limited by the closure, not the quadrature. *Right:* relative score error against $1 - \cos$ of the angle between projected and reference score vectors; directional error is one to two orders of magnitude smaller than magnitude error at every t .

Chapter 6

Learning the Parameters: EM on the Chain

Everything so far assumed the transition kernel was given. This chapter drops that assumption: the kernel is unknown and is recovered from the noised sequences alone, with no clean data, no supervision, and no differentiation through the inference. The tool is expectation–maximisation, and the reason it is cheap here is Fisher’s identity: at each parameter value one forward–backward pass of belief propagation already contains the exact gradient of the marginal log-likelihood, so every E-step is the recursion of Chapter 4 with the kernel held fixed, and nothing is ever back-propagated through it.

6.1 Model: an estimated mixture innovation

The clean chain is (5.1) in the index convention of this chapter, $a_1 \sim \mathcal{N}(0, 1)$ and $a_i = \alpha a_{i-1} + \varepsilon_i$ for $i = 2, \dots, L$, with i.i.d. centred innovations of variance $\sigma_\eta^2 = 1 - \alpha^2$ now drawn from a member of a parametric family; noising is the channel (3.4). The estimated family is a C -component Gaussian mixture on the innovation,

$$K_\theta(a' | a) = \sum_{c=1}^C \pi_c \mathcal{N}(a' - \alpha a; \nu_c, s_c^2), \quad \theta = \{\alpha, (\pi_c, \nu_c, s_c)_{c=1}^C\}, \quad (6.1)$$

with $\pi_c \geq 0$, $\sum_c \pi_c = 1$, $s_c^2 \geq s_{\min}^2$, $\nu_c \in \mathbb{R}$ and $\alpha \in (-1, 1)$, so that both the autoregressive coefficient and the innovation shape are estimated jointly by maximum marginal

likelihood from noised data; the count of free parameters is $3C$, and $C = 1$ is the Gaussian chain of Chapter 3. The fitted family deliberately enforces neither $\sum_c \pi_c \nu_c = 0$ nor a matched total variance, although the generating innovations satisfy both: projecting the means back to zero after each update broke monotone ascent by $\sim 10^{-2}$ nats per edge when tried, whereas the unconstrained update is exactly monotone and the fitted innovation mean lands at the 10^{-3} level on its own, so the moment conditions are reported as diagnostics rather than imposed. The constraint $|\alpha| < 1$ never binds in any fit reported here, and component labels are exchangeable, so label-dependent diagnostics use a fixed ordering.

6.2 The EM algorithm on the chain

Observed is $x = (x_1, \dots, x_L)$; latent are the clean trajectory a and, for every transition, the label $c_i \in \{1, \dots, C\}$ of the component that generated its innovation. Maximum likelihood targets $\mathcal{L}(\theta) = \log p_\theta(x) = \log \sum_c \int p_\theta(a, c, x) da$, a logarithm outside an integral over the trajectory and a sum over labels. With $q^{(k)}(a, c) := p_{\theta^{(k)}}(a, c | x)$ the posterior at the current iterate and the surrogate $Q(\theta | \theta^{(k)}) := \mathbb{E}_{q^{(k)}}[\log p_\theta(a, c, x)]$, taking $\mathbb{E}_{q^{(k)}}$ of $\log p_\theta(x) = \log p_\theta(a, c, x) - \log p_\theta(a, c | x)$ gives the exact decomposition

$$\mathcal{L}(\theta) = Q(\theta | \theta^{(k)}) + \mathcal{H}(q^{(k)}) + \text{KL}(q^{(k)} \| p_\theta(a, c | x)), \quad (6.2)$$

a lower bound $Q + \mathcal{H}$ that touches $\mathcal{L}$ at $\theta^{(k)}$, so any $\theta^{(k+1)}$ that increases Q increases $\mathcal{L}$. The E-step computes $q^{(k)}$; the M-step increases Q . Because the posterior factor graph is a tree, one forward and one backward sweep return the exact single-site and pairwise posterior marginals at the level of functional messages: the E-step needs neither a variational approximation nor Monte Carlo, and is exact under the model before the grid is introduced.

Conditional on $c_i = c$ the transition is a single Gaussian, $\tilde{K}_\theta(a', c | a) = \pi_c \mathcal{N}(a' - \alpha a; \nu_c, s_c^2)$, and only the transition factors of the complete joint law $p(a_1) \prod_i \tilde{K}_\theta(a_i, c_i |$

$a_{i-1}) \prod_i p(x_i | a_i)$ depend on θ :

$$\log p_\theta(a, c, x) \stackrel{c}{=} \sum_{i=2}^L \sum_{c=1}^C \mathbf{1}\{c_i = c\} \left[\log \pi_c - \frac{1}{2} \log s_c^2 - \frac{(a_i - \alpha a_{i-1} - \nu_c)^2}{2s_c^2} \right], \quad (6.3)$$

$\stackrel{c}{=}$ denoting equality up to θ -independent terms. Exposing the label turns the log of a mixture into a weighted sum of Gaussian log-densities with closed-form maximisers. Its expectation needs, for each edge, only the pairwise posterior supplied by the two sweeps,

$$b_i(a_{i-1}, a_i) \propto \vec{m}_{i-1}(a_{i-1}) \phi_{i-1}(a_{i-1}) K_{\theta^{(k)}}(a_i | a_{i-1}) \phi_i(a_i) \overleftarrow{m}_i(a_i), \quad \phi_i(a_i) = p(x_i | a_i), \quad (6.4)$$

and the responsibility $r_{i,c} = \mathbb{P}_{\theta^{(k)}}(c_i = c | a_{i-1}, a_i)$ of component c given the pair, Bayes' rule on the finite mixture at the current parameters, which needs no further recursion. Writing $\langle \cdot \rangle_i$ for expectation under b_i , the E-step reduces to six sufficient statistics per component, summed over all edges of all sequences,

$$\begin{aligned} R_c &= \sum_i \langle r_{i,c} \rangle_i, & U_c &= \sum_i \langle r_{i,c} a_i \rangle_i, & V_c &= \sum_i \langle r_{i,c} a_{i-1} \rangle_i, \\ P_c &= \sum_i \langle r_{i,c} a_{i-1} a_i \rangle_i, & W_c &= \sum_i \langle r_{i,c} a_{i-1}^2 \rangle_i, & G_c &= \sum_i \langle r_{i,c} a_i^2 \rangle_i, \end{aligned} \quad (6.5)$$

in terms of which, with $S_c(\alpha, \nu_c) := G_c - 2\alpha P_c + \alpha^2 W_c - 2\nu_c U_c + 2\alpha \nu_c V_c + \nu_c^2 R_c$,

$$Q(\theta | \theta^{(k)}) \stackrel{c}{=} \sum_{c=1}^C \left[R_c \log \pi_c - \frac{R_c}{2} \log s_c^2 - \frac{S_c(\alpha, \nu_c)}{2s_c^2} \right]. \quad (6.6)$$

Fisher's identity. Differentiating $\log p_\theta(x)$ under the integral, licit because the Gaussian factors and the mixture kernel make $\nabla_\theta p_\theta(a, c, x)$ dominated on the compact parameter set every fit stays in,

$$\nabla_\theta \mathcal{L}(\theta) = \mathbb{E}_{a,c|x,\theta} [\nabla_\theta \log p_\theta(a, c, x)] = \mathbb{E}_{a,c|x,\theta} \left[\sum_{i=2}^L \nabla_\theta \log \tilde{K}_\theta(a_i, c_i | a_{i-1}) \right], \quad (6.7)$$

so that $\nabla_{\theta} \mathcal{L}(\theta)|_{\theta^{(k)}} = \nabla_{\theta} Q(\theta | \theta^{(k)})|_{\theta^{(k)}}$. BP is used only to compute the posterior; once computed it is held fixed, and the derivative acts on the candidate kernel inside $\log \tilde{K}_{\theta}$, not on the messages that produced the posterior. BP supplies posterior expectations, Fisher's identity converts them into the likelihood gradient, and no derivative is taken through the inference.

M-step. Setting the derivatives of (6.6) to zero block by block, with a multiplier for $\sum_c \pi_c = 1$,

$$\boxed{\pi_c^+ = \frac{R_c}{\sum_d R_d}, \quad \nu_c^+ = \frac{U_c - \alpha V_c}{R_c}, \quad (s_c^2)^+ = \frac{G_c - 2\alpha P_c + \alpha^2 W_c - R_c(\nu_c^+)^2}{R_c}}, \quad (6.8)$$

and, with the mixture block fixed,

$$\boxed{\alpha^+ = \frac{\sum_c s_c^{-2} (P_c - \nu_c V_c)}{\sum_c s_c^{-2} W_c}}, \quad (6.9)$$

a precision-weighted regression in which components with smaller innovation variance carry more information about α ; at $C = 1$, $\nu_1 = 0$ it is the posterior analogue of the least-squares slope, $\sum_i \mathbb{E}[a_{i-1} a_i | x] / \sum_i \mathbb{E}[a_{i-1}^2 | x]$. The iteration is

$$\boxed{\text{E-step: } \theta^{(k)} \xrightarrow{\text{BP}} \{R_c, U_c, V_c, P_c, W_c, G_c\}_{c=1}^C \quad \text{M-step: } \{R_c, \dots, G_c\} \xrightarrow{(6.8), (6.9)} \theta^{(k+1)}}. \quad (6.10)$$

The updates are conditional maximisers, so the implementation is an ECM scheme (Meng and Rubin, 1993); by (6.2) any increase of Q increases $\mathcal{L}$, and the two blocks are alternated until the gain in Q per edge falls below tolerance or sixteen inner sweeps are reached. The cap is chosen from the numerics: at $C = 8$, $N = 256$ and 40 outer iterations, raising it from four to sixteen moves the fitted excess kurtosis from 1.68 to 2.14 while $\hat{\alpha}$ changes in the fourth decimal, so the shape converges far more slowly than the correlation, and by roughly 60 outer iterations the inner budget has little remaining effect. Two guards keep (6.8) defined: the responsibility mass is floored at

10^{-300} before division and the variance at $s_c^2 \geq 10^{-6}$, far below the grid spacing, so the floor marks degeneracy rather than shaping a resolved fit; a separate reporting gate (Appendix A) excludes fits whose narrowest component is under two grid cells wide. The derivation concerns the continuum model; the implementation evaluates densities on the grid of Section 6.3, so the objective whose monotone increase Table 6.2 certifies is the quadrature-defined surrogate of the discretised factor model.

6.3 The numerical representation

As in Section 5.3, messages live on N_g equispaced points of $[-A, A]$ with trapezoidal integrals; the pairwise belief is an $N_g \times N_g$ array, and one EM iteration costs $O(LN_g^2)$ per sequence with a factor C only in the responsibilities. Two numerical errors are kept apart (Table 6.1, Figure 6.1). *Truncation*, from replacing $\mathbb{R}$ by $[-A, A]$, is monitored through the forward-message boundary mass and, because it depends on the sampled trajectory, reported over 256 chains, whose batch maximum exceeds a single-chain reading by a median factor 86 and by as much as 3,281. *Quadrature*, from the spacing $h = 2A/(N_g - 1)$, is $O(h^2)$ in the interior, the residual falling by a factor 4.00 per halving, and does not converge near the boundary because the remaining error there is truncation. The transition array is the continuum density evaluated on the grid and integrated with trapezoidal weights *without* renormalising each parent-conditioned column: BP is then exact for the resulting discretised factor model, which is not exactly a normalised finite-state kernel, and the discrepancy is the column-mass residual of Table 6.1. Column renormalisation would introduce a θ -dependent normaliser Z_θ whose gradient would enter Fisher’s identity and destroy the closed-form M-step, so the convention fits the continuum kernel through the quadrature rule, with truncation controlled separately. Posterior-mean errors are amplified in the score by $e^{-t}/\Delta_t \sim 1/2t$, so numerical and estimation deficiencies are most visible at small t .

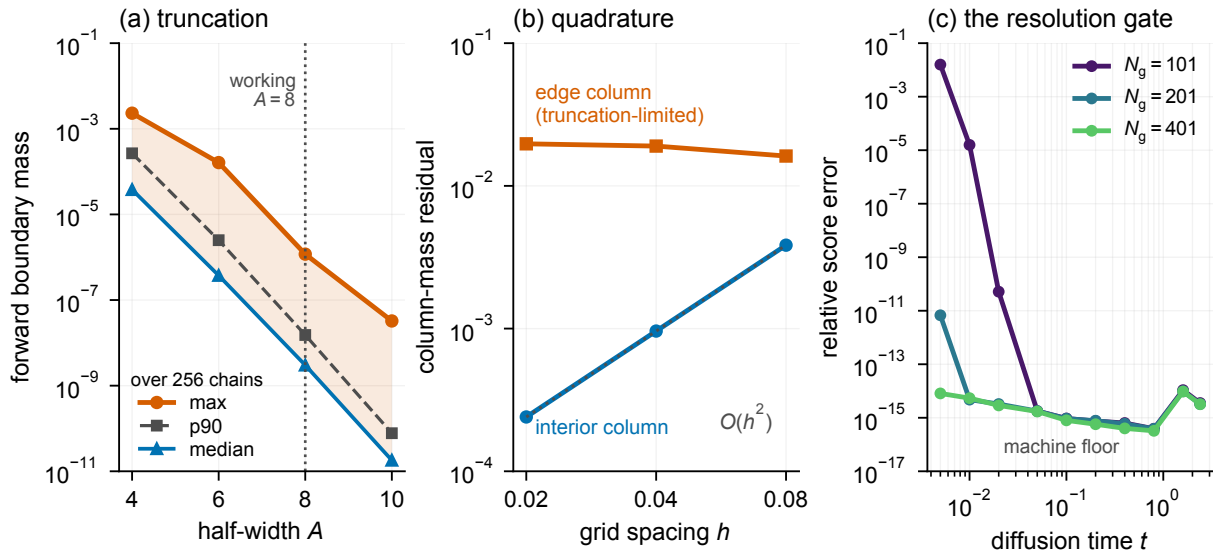


Figure 6.1: (a) Boundary mass falls rapidly with the half-width A but varies substantially across chains. (b) Interior column-mass error follows the $O(h^2)$ rate; the edge error is truncation-dominated. (c) Relative score error against diffusion time for three resolutions; failure sets in where (5.8) is violated.

Table 6.1: Numerical error diagnostics. *Upper:* boundary mass against half-width at $N_g = 401$, over 256 chains. *Lower:* column-mass residual against spacing at $A = 8$. Working choice: $N_g = 401$, $A = 8$.

<i>truncation: boundary mass against half-width ($N_g = 401$)</i>				
half-width A	4	6	8	10
max over chains	2.3×10^{-3}	1.6×10^{-4}	1.2×10^{-6}	3.2×10^{-8}
90th percentile	2.7×10^{-4}	2.5×10^{-6}	1.5×10^{-8}	7.8×10^{-11}
median	3.8×10^{-5}	3.7×10^{-7}	3.0×10^{-9}	1.8×10^{-11}
max / median	61×	434×	392×	1796×
<i>quadrature: column-mass residual against spacing ($A = 8$)</i>				
N_g / h	201 / 0.080	401 / 0.040	801 / 0.020	rate
interior column	3.84×10^{-3}	9.61×10^{-4}	2.40×10^{-4}	4.00×
edge column	1.62×10^{-2}	1.90×10^{-2}	1.97×10^{-2}	per halving none

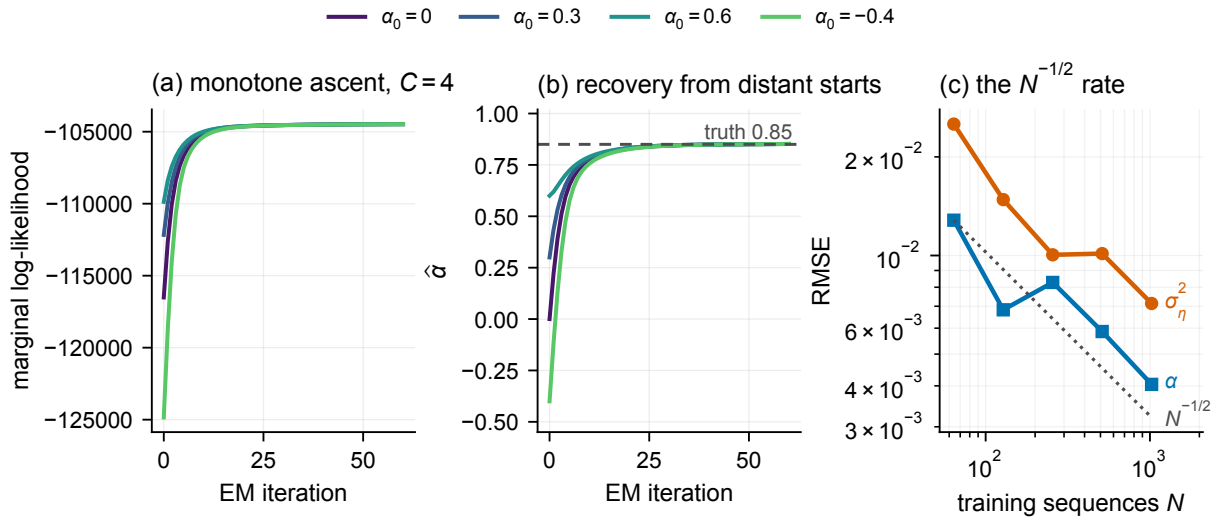


Figure 6.2: Diagnostics at $C = 4$. (a) Marginal log-likelihood increases monotonically from four initialisations of α . (b) $\hat{\alpha}$ converges to $[0.8517, 0.8520]$ from starts spanning $[-0.4, 0.6]$, against the true 0.85. (c) RMSE of $\hat{\alpha}$ and of the innovation variance against N over 16 seeds, with an $N^{-1/2}$ reference.

6.4 Diagnostics, and what EM recovers

The recursion agrees with independent references to machine precision and the surrogate never decreases in any logged run (Table 6.2); $\hat{\alpha}$ converges to the same value from initialisations spanning $[-0.4, 0.6]$, so it is estimated rather than supplied, and its RMSE and that of the innovation variance decay approximately at the $N^{-1/2}$ rate (Figure 6.2).

Table 6.2: Correctness and monotonicity checks at $N_g = 401$, $A = 8$.

check	reference	discrepancy
posterior mean	closed-form Gaussian chain (Chapter 3)	9.2×10^{-15}
posterior mean	brute-force enumeration, independent code	1.6×10^{-14}
log-evidence	brute-force enumeration, independent code	1.8×10^{-15}
min. increase of Q , twelve steps	Gaussian kernel	+2.61
	Laplace kernel	+4.34
	$C = 8$ mixture kernel	+4.18
decreases in any logged run		0

The parameters are not recovered equally well (Table 6.3, Figure 6.3): at $C = 8$ and

$N \geq 512$ the correlation is recovered to below 0.1% relative error, the innovation variance to the percent level, and the fourth cumulant only to tens of percent. Each row is a single fit and the shape column is not monotone in N , the seed-to-seed spread of the fourth cumulant being of the order of its error, so the rows demonstrate the hierarchy, not a convergence law for the shape. The ordering is what the channel predicts: by (5.3) the fourth cumulant is attenuated as e^{-4t} while the second moment is preserved, and a finite Gaussian mixture also rounds the Laplace cusp, so the residual shape error mixes information lost through the channel with approximation by the family. “Recovery” means numerical agreement over the studied regime; no identifiability or consistency theorem is claimed for the mixture family. The estimated kernel is therefore most reliable in its correlation structure, reasonably accurate in scale, and substantially less informative about tail shape.

Table 6.3: Estimates after 61 EM iterations on noised Laplace chains ($\alpha = 0.85$, $\sigma_\eta^2 = 0.2775$, excess kurtosis 3.0, $L = 32$).

C	N	$\hat{\alpha}$	$\widehat{\sigma_\eta^2}$	$\hat{\kappa}_4$	$ \Delta\alpha /\alpha$	$ \Delta\sigma_\eta^2 /\sigma_\eta^2$	$ \Delta\kappa_4 /\kappa_4$
4	128	0.8649	0.2686	1.61	1.8%	3.2%	46%
4	512	0.8518	0.2844	2.27	0.2%	2.5%	24%
4	2048	0.8465	0.2818	1.82	0.4%	1.5%	39%
8	128	0.8294	0.2824	3.18	2.4%	1.8%	6%
8	512	0.8503	0.2694	2.26	0.04%	2.9%	25%
8	2048	0.8507	0.2793	2.72	0.08%	0.7%	9%

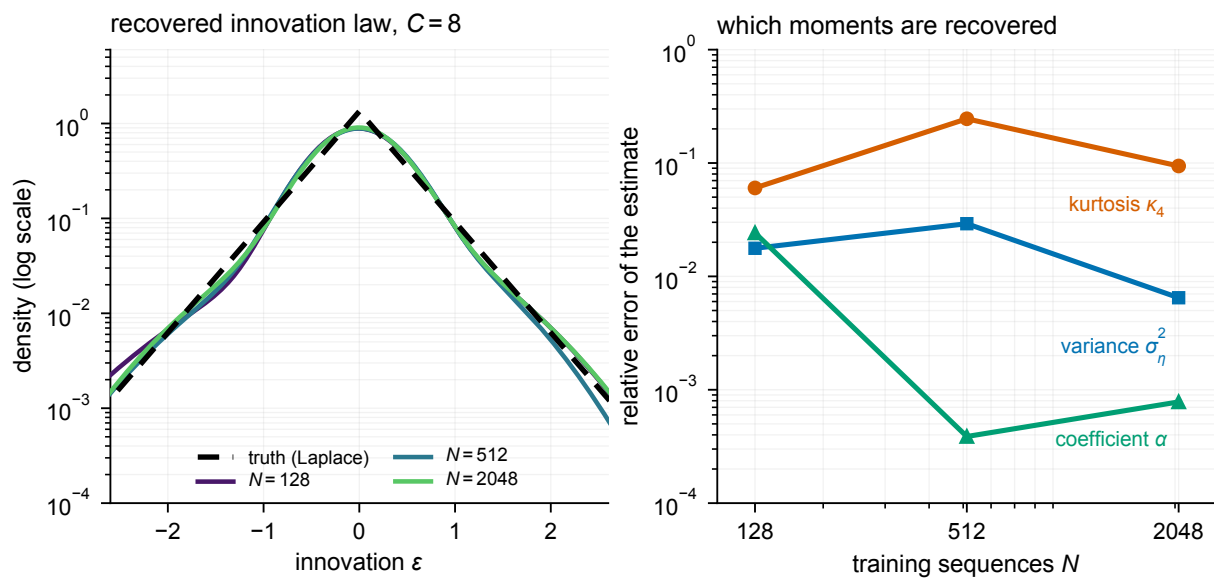


Figure 6.3: *Left:* fitted $C = 8$ mixture innovation densities for three data budgets against the Laplace truth; the discrepancies sit at the cusp and in the tails. *Right:* relative errors in α , innovation variance and fourth cumulant against N , the hierarchy from correlation to scale to shape.

Chapter 7

Learning the Kernel, and What Convergence Costs

Chapter 6 established that the estimator is correct: the E-step is exact, the surrogate ascends, and the recovery error of the second-order coordinates falls approximately as $N^{-1/2}$ over the tested range. None of that says how long it takes. Correctness is a statement about the fixed point; this chapter is about the map that reaches it, and the two are independent: an estimator can be exactly right in the limit and still be routinely reported before it arrives. The finding is that the map does not converge at one speed. The autoregressive coefficient and the innovation *shape* contract at rates that differ by a factor of eight in the exponent, so a stopping rule calibrated on the first certifies a kernel that has not converged in the second, and every non-Gaussian claim in this thesis depends on the coordinate that settles last.

7.1 Why there are two rates

The M-step defines a map $\theta^{(k+1)} = \mathcal{M}(\theta^{(k)})$ whose fixed points are stationary points θ^* of the marginal likelihood, and EM converges linearly, at a rate set by the spectrum of the Jacobian of that map at the fixed point, which [Dempster et al. \(1977\)](#) identified. Expanding about θ^* ,

$$\theta^{(k+1)} - \theta^* = J(\theta^*) (\theta^{(k)} - \theta^*) + O(\|\theta^{(k)} - \theta^*\|^2), \quad J = \left. \frac{\partial \mathcal{M}}{\partial \theta} \right|_{\theta^*}. \quad (7.1)$$

With $I_{\text{com}} = -\mathbb{E}[\nabla^2 \log p_\theta(a, c, x) \mid x]$ the expected complete-data information given x , $I_{\text{obs}} = -\nabla^2 \mathcal{L}$ the observed information and I_{mis} their difference, all at θ^* ,

$$I_{\text{com}} = I_{\text{obs}} + I_{\text{mis}}, \quad J(\theta^*) = I_{\text{com}}^{-1} I_{\text{mis}} = \mathbb{I} - I_{\text{com}}^{-1} I_{\text{obs}} : \quad (7.2)$$

the rate of EM is the fraction of the information that the latent variables carry away: as $I_{\text{mis}} \rightarrow I_{\text{com}}$ the largest eigenvalue of J approaches one and progress per update approaches nothing. One scope condition governs everything measured below. Identity (7.2) is the Jacobian of *exact* EM; the mixture M-step of Chapter 6 cycles conditional maximisations with at most sixteen inner sweeps, and with r sweeps its Jacobian is $G^r + (\mathbb{I} - G^r)J$, G being the block Gauss–Seidel matrix of those maximisations. So (7.2) is an ordering heuristic for which coordinate should be slow, never a calibrated prediction of how slow.

A coordinate aligned with an eigenvector of J with eigenvalue λ contracts as λ^k , so stopping it when its relative distance to the limit first falls below a tolerance τ takes

$$k(\tau, \lambda) \approx \frac{\log \tau}{\log \lambda}, \quad \frac{k_2}{k_1} \approx \frac{\log \tau_2}{\log \tau_1} \cdot \frac{\log \lambda_1}{\log \lambda_2}, \quad (7.3)$$

which separates a choice of tolerance from a property of the model. Two limits are stated now and verified below: it is an asymptotic-tail statement, a trace can fall much faster than its terminal slope during the transient, so (7.3) upper-bounds a measured settling time; and it speaks about eigenmodes, whereas a parameter coordinate mixes several, so a slope fitted to a coordinate’s trace is an empirical contraction factor, identified with an eigenvalue only if one mode dominates its window.

The observation channel enters (7.2) twice. *On the rate*: fitting on clean transition pairs leaves only the mixture label latent; passing the chains through the channel makes the whole state a latent as well, so I_{mis} grows by everything the posterior over a does not resolve, J moves towards the identity, and the map slows, a statement about iterations that would persist at infinite sample size. *On the accuracy*: the channel also shrinks I_{obs} , and the asymptotic covariance of the estimator is I_{obs}^{-1}/N , a statement about data

that iterating longer does not remove. Section 7.4 measures it.

Which coordinate should be slower the model says before any fit is run. The channel is variance preserving and the chain is normalised to unit marginal variance, so the second cumulant of the noised variable is preserved exactly at every t while the fourth is damped,

$$\kappa_2(X_t) = \kappa_2(A) \quad \text{for all } t, \quad \kappa_4(X_t) = e^{-4t} \kappa_4(A) \quad (7.4)$$

(5.3). The autoregressive coefficient is a second-order quantity, identified by the lag-one covariance, which the channel attenuates in a known, invertible way; the innovation shape is a fourth-order quantity, and the channel removes an e^{-4t} fraction of exactly the structure that identifies it. The information the channel takes away is concentrated on the shape coordinate, which by (7.2) is the coordinate whose eigenvalue of J is pushed closest to one. *Prediction:* $\lambda_{\kappa_4} > \lambda_\alpha$, and by (7.3) a settling time for the shape that is a large multiple of the settling time for the correlation.

7.2 Measuring the two rates

A dedicated sweep fits the mixture kernel through the channel at $N = 512$, $C = 8$ and the uniform noise design, over 16 seeds run to 800 updates, recording the full parameter trace. Figure 7.1(a) plots each coordinate's relative distance to its own end-of-run value: both are straight on a log axis over a wide window, which is (7.1) being obeyed, and the slopes differ sharply (Table 7.1).

Table 7.1: Empirical coordinate-wise contraction factors λ read from the traces of Figure 7.1(a): median over 16 seeds of the least-squares slope of $\log |\theta^{(k)} - \theta^{(800)}|$ over the fitting window. A coordinate mixes eigenmodes of the ECM map and the excess kurtosis is a nonlinear functional of the mixture parameters, so these slopes are not Jacobian eigenvalues. The asymptotic column applies (7.3) with the tolerance actually used; the last column is the measured median.

coordinate	fitting window	λ	rel. tolerance τ	asymptotic $k(\tau, \lambda)$	measured median
α	20–150	0.9686	1.2×10^{-3}	211	80
σ_η^2	20–150	0.9671	—	—	68
κ_4	60–400	0.9960	6.7×10^{-3}	1256	229
ratio κ_4/α		log-ratio 8.0		5.9	2.8

The prediction is confirmed in direction and in magnitude of effect. The shape contracts by 0.9960 per update against 0.9686 for the correlation, a factor of 8.0 in the exponent, and the innovation *variance* settles with the correlation at 0.9671, the same statement from the other side: it is a second-order quantity and the channel preserves it. Read through (7.2) this ordering is consistent with a slow, shape-dominated eigenmode whose missing information sits far closer to one than the correlation’s; consistent with, not a measurement of, since turning slopes into eigenvalues would require the Jacobian of the implemented map, which is not computed. Where the linear theory is not quantitative is the constant: (7.3) predicts a settling ratio of 5.9, the paired measurement gives 2.8, and the asymptotic counts over-predict both individual settling times by a factor of two to three, because both curves fall faster in their first hundred updates than their fitted slopes, so both tolerances are crossed in the transient rather than in the asymptotic regime. The eigenvalues bound the tail, not the crossing of a threshold.

7.3 Settling times and the stopping rule

Across 16 seeds the correlation settles at a median of 80 updates and the shape at 229, the largest observed value being 638 (Figure 7.1(b)); the ratio formed within each fit has median 2.8, against 2.9 for the quotient of medians, and the paired figure is the

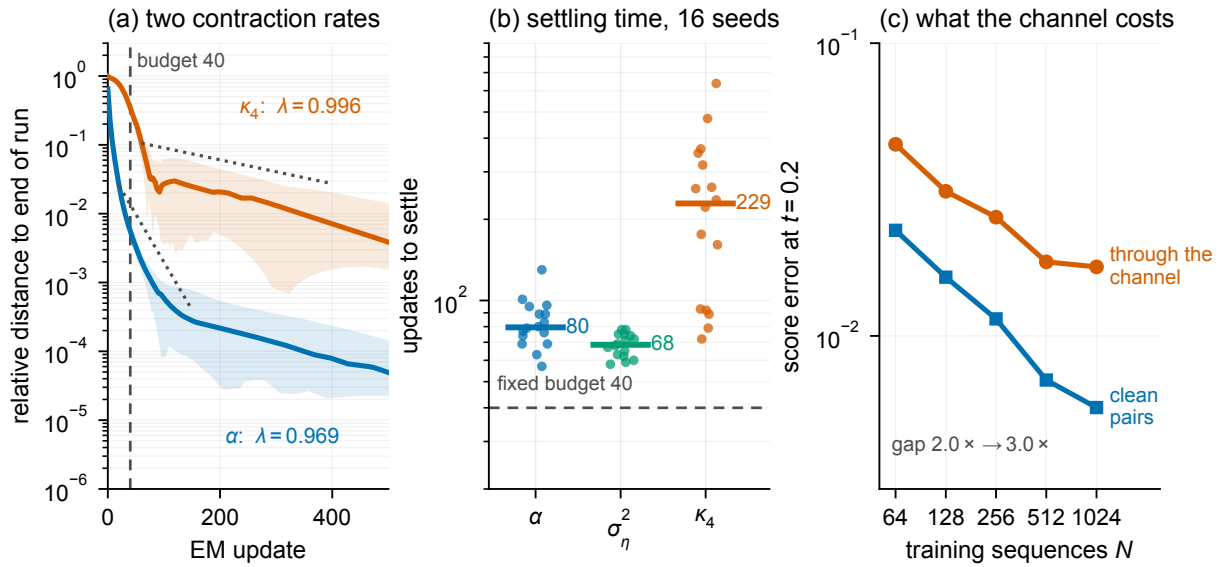


Figure 7.1: (a) Relative distance of each coordinate to its end-of-run value, median over 16 seeds with the 10–90% band; dotted lines are the fitted rates of Table 7.1. (b) Settling times per seed, with medians: correlation and innovation variance settle together near 80 updates, the shape at 229 with a tail to 638; the dashed line is the fixed budget of 40 updates several experiments used as a default, below every observed settling time of every coordinate. (c) The channel’s accuracy cost, which is not a rate: relative score error at $t = 0.2$ of a kernel fitted on clean transition pairs against one fitted on the same chains through the channel, both run to convergence; the gap widens with data rather than closing.

one reported. “Settled” means trace stability, the first update after which a coordinate stays within 10^{-3} of its end-of-run value for α or 2×10^{-2} for the excess kurtosis, not distance to an optimiser, and the medians come from the 800-update shard, the only run length at which no configuration settles near its own cap. Against them, the fixed budget of 40 iterations that several experiments in Chapter 8 used as a default falls short of the shape’s median by about a factor of six and leaves 105 of the sweep’s 112 configurations, 94%, unsettled, a count that pools both shards and is conservative because censoring can only shorten a measured settling time. Table 7.2 makes the same point on one configuration: the correlation is essentially in place by thirty updates while the innovation shape is still climbing at a hundred and twenty. The mixture label is a latent variable of its own and the channel compounds the missing information, the mechanism of Dempster et al. (1977) appearing in the convergence rate exactly where (7.2) places it.

Table 7.2: What the observation channel costs in rate rather than in where the fit lands. Above: the through-channel fit at $N = 512$, $C = 8$, as a median over 16 seeds at four updates — the correlation is essentially in place by 30 while the innovation shape is still moving at 120. Below: where the two arms finish, fit on clean transition pairs and on the same chains once through the channel. Generating values are $\alpha = 0.85$ and excess kurtosis 3.0; a finite mixture does not contain the generating law exactly, so what is approached is the best fit within the family.

update	α	innovation excess kurtosis
1	0.4402	0.078
30	0.8412	1.275
120	0.8499	3.019
600	0.8504	2.980
<i>where each arm finishes</i>		
clean transition pairs	0.8509	2.939
through the channel	0.8526	2.729

7.4 What the channel costs in accuracy at equal data

The endpoints in Table 7.2 are close: the two arms finish within 0.002 of each other on the correlation and within a fifth of a unit on excess kurtosis, both near the generating values. Under a correctly specified, identifiable model the clean and noised likelihoods share their population maximiser and shared endpoints are what shrinking I_{obs} predicts; the fitted family here is a finite mixture that does not contain the Laplace innovation exactly, and under misspecification the two pseudo-true optima need not coincide, so the close endpoints are a measured approximate invariance for this family in this regime, not a theorem. Either way they are not evidence that the channel is free: the second effect lives in the variance, and Table 7.3 measures it on the quantity the thesis uses, the relative score error of the fitted kernel at $t = 0.2$.

Table 7.3: Relative score error at $t = 0.2$ of the fitted kernel, clean transition pairs against the same chains through the channel, $C = 8$, mean over three replicates, an exploratory design that carries no usable uncertainty estimate, so the row-to-row shape of either curve should not be over-read.

N	64	128	256	512	1024
clean transition pairs	0.0229	0.0159	0.0114	0.0071	0.0057
through the channel	0.0451	0.0312	0.0254	0.0179	0.0172
ratio	$2.0\times$	$2.0\times$	$2.2\times$	$2.5\times$	$3.0\times$

The penalty is real at every size and over the measured range it grows: the clean arm improves fourfold, close to the $N^{-1/2}$ behaviour of Figure 6.2(c), while the noised arm improves only 2.6-fold and is nearly flat between $N = 512$ and 1024, so the ratio moves from $2.0\times$ to $3.0\times$. Under a regular, correctly specified model both arms obey $\sqrt{N}(\hat{\theta} - \theta^*) \Rightarrow \mathcal{N}(0, I_{\text{obs}}^{-1})$, so a smaller observed information enlarges the constant of an $N^{-1/2}$ law and cannot by itself produce a plateau; a genuine flattening needs something else, the finite-mixture approximation floor, quadrature and stopping tolerances, a shift of the noised pseudo-true target under misspecification, or a pre-asymptotic range, and three replicates over five sizes cannot separate these. What the table establishes is a larger finite-data cost through the channel, widening over the tested range, not a changed asymptotic rate; reduced I_{obs} remains the mechanism for the level of the penalty, and equal endpoints with unequal precision are the two halves of one statement.

Chapter 8

What Structure Buys, and Where It Stops Paying

Chapter 6 built an estimator exact in its inference step and approximate only in its numerical representation and its budget; Chapter 7 measured what it costs to run. This chapter asks what it is *worth*, relative to networks trained on the same clean trajectories and given none of the structure: how much supplying the chain reduces denoising error at a fixed number of clean trajectories, how much of that survives against a baseline given locality but not the model, how much mixture capacity the estimator needs, and how far the Markov assumption can be wrong before the structure stops paying.

8.1 What is compared, and how it is scored

Fix a diffusion time t and a test set $\{x^{(j)}\}_{j=1}^J$ of noised sequences from the model of Section 6.1, and let S^* be the exact joint score, supplied by the validated grid reference under the true kernel. For an estimator $\hat{S}$ the relative L^2 score error is

$$\mathcal{E}[\hat{S}; t] = \left(\frac{\sum_j \left\| \hat{S}(x^{(j)}, t) - S^*(x^{(j)}, t) \right\|^2}{\sum_j \|S^*(x^{(j)}, t)\|^2} \right)^{1/2}, \quad (8.1)$$

scale free, with 0.1 meaning a tenth of the signal’s own size. Pooling over the noise schedule $\mathcal{T}$ pools the two sums, not the ratios,

$$\mathcal{E}^2[\hat{S}] = \sum_{t \in \mathcal{T}} w_t \mathcal{E}^2[\hat{S}; t], \quad w_t = \frac{\mathbb{E} \|S^*(X_t, t)\|^2}{\sum_{s \in \mathcal{T}} \mathbb{E} \|S^*(X_s, s)\|^2}, \quad (8.2)$$

the weights being the measured reference score energies of the frozen test bundle. The covariance-matched Gaussian chain explains their shape: there $\mathbb{E} \|S^*\|^2 = \text{tr} \Sigma_t^{-1}$, which falls from $\text{tr} Q_0 = 193.4$ at $t \rightarrow 0$ to $L = 32$ as $t \rightarrow \infty$ with no $1/t$ singularity, so the schedule weights its lower-noise levels most heavily; the four smallest of the twelve levels carry 54.5% of the measured weight (Table 8.5).

The two arms are the EM–BP estimator, given the Markov factorisation, and a denoising-score-matching network trained by (2.10) with no structural information. Both are fitted from the same N clean trajectories, and that, not “the same data”, is the equality enforced. What each arm sees of those trajectories differs, and the difference runs *against* the result reported here: EM–BP sees one frozen noised realisation of each trajectory and never the clean signal, whereas the network is trained on paired (a, x) data as (2.10) requires and redraws t and the corrupting noise at every gradient step. Both asymmetries are deliberate, the baseline being the standard training recipe for its class. The reported quantity is

$$R = \frac{\mathcal{E}_{\text{base}}}{\mathcal{E}_{\text{EM}}}, \quad R > 1 \text{ meaning the structured estimator is more accurate,} \quad (8.3)$$

formed cell by cell, a cell being one (training-set size, seed) pair with both arms refitted: a statement about value per clean trajectory under the two standard recipes, not about data efficiency at a matched information interface. Because R is a ratio of the quantities (8.1), the aggregation matters: squared error and reference energy are summed over the twelve levels before the square root, per seed, the ratio is formed within the seed, and the reported number is the mean of the sixteen paired per-seed ratios with its standard error, the seed being the inferential unit. Appendix A recomputes the same

cells under alternative summaries: the direction and order of magnitude are the same in all, but the mean of per-level ratios, which earlier drafts reported, is systematically the largest ($7.0\text{--}17.6\times$ against the pooled $6.6\text{--}10.9\times$). Table 8.1 fixes the protocol shared by Sections 8.3–8.5; Section 8.6 changes the prior and Section 8.7 is a likelihood calculation.

Table 8.1: The protocol shared by Sections 8.3–8.5. The generating prior is a Laplace chain, so the reference score is genuinely nonlinear. Validation, test and training bundles are disjoint, and no arm sees the test bundle before its budget is chosen.

generating prior	Laplace AR(1), $\alpha = 0.85$, unit marginal variance
innovation law	$\eta \sim \text{Laplace}(0, b)$, $b = \sqrt{\sigma_\eta^2/2}$, excess kurtosis +3
chain length	$L = 32$ sites
channel	variance preserving, $x = e^{-t}a + \sqrt{\Delta_t}\varepsilon$
noise schedule	12 levels, geometric, $t \in [0.05, 3.0]$, ratio 1.45
reference score	grid BP under the true kernel, $N_g = 401$, $A = 8$
fitted kernel	$C = 8$ mixture components, 24 free parameters
training sizes	$N \in \{32, \dots, 4096\}$ (headline) or $\{32, 128, 512, 2048\}$
replicates	16 seeds, both arms refitted per cell
validation / test	1,024 / 2,048 sequences (structured runs); 256 / 256 (headline)

8.2 The estimators

Every network arm is read as a score. Predicting the noise or the clean signal are two parameterisations of one object,

$$\hat{S}_\varepsilon(x, t) = -\frac{\hat{\varepsilon}_\phi(x, t)}{\sqrt{\Delta_t}}, \quad \hat{S}_a(x, t) = \frac{e^{-t}\hat{a}_\phi(x, t) - x}{\Delta_t}, \quad (8.4)$$

identical as function classes but different as optimisation problems, since the substitution rescales the residual by $e^t\sqrt{\Delta_t}$ and reweights the noise levels; both are offered to every arm and chosen on validation, and the screen below settles it in favour of the noise head within every architecture. Throughout, $\tau(t) = (t, e^{-t}, \sqrt{\Delta_t})$ is a time embedding broadcast to every site.

EM–BP fits the mixture kernel (6.1) by Section 6.2 and evaluates the score through the score–posterior identity, with $3C = 24$ free parameters independent of L ; it is supplied the Markov factorisation, the linear-plus-innovation form of the kernel and homogeneity across sites, and estimates the innovation law and α . The *unstructured denoiser* is a fully connected tanh network mapping the whole sequence and $\tau(t)$ to the whole output,

$$f_\phi(x, t) = W_3 \tanh(W_2 \tanh(W_1[x; \tau(t)] + b_1) + b_2) + b_3, \quad (8.5)$$

hidden width 128, 25,248 parameters, a thousand times EM–BP’s; nothing in it knows the sites are ordered, since permuting the coordinates of x and of the output is absorbed by permuting columns of W_1 and rows of W_3 . Because a large part of any margin over (8.5) could be permutation-blindness rather than the Markov model, three architectures that encode sequence structure are screened. The *weight-shared window head* of radius r applies one two-layer tanh predictor g_ϕ of width W at every site to its window,

$$\hat{y}_k = g_\phi(x_{k-r}, \dots, x_{k+r}; \tau(t)), \quad k = 1, \dots, L, \quad (8.6)$$

a one-dimensional convolution of width $2r + 1$ with pointwise layers, with a parameter count $(2r + 4)W + W^2 + 3W + 1$ independent of L and a receptive radius fixed at r . The *dilated convolutional stack* is a residual stack of three-tap convolutions at dilations $d_\ell \in \{1, 2, 4, 8\}$,

$$h_k^{(\ell)} = h_k^{(\ell-1)} + P_\ell \tanh\left(\sum_{s \in \{-d_\ell, 0, d_\ell\}} W_{\ell,s} h_{k+s}^{(\ell-1)} + b_\ell\right), \quad (8.7)$$

with receptive radius 15, spanning the chain. The *bidirectional message-passing* network runs forward and backward recurrences with shared local updates and a joint

readout,

$$\vec{h}_k = \tanh(Az_k + F\vec{h}_{k-1} + a), \quad \overleftarrow{h}_k = \tanh(Bz_k + G\overleftarrow{h}_{k+1} + b), \quad \hat{y}_k = C[\vec{h}_k; \overleftarrow{h}_k; z_k] + c, \quad (8.8)$$

with $z_k = [x_k; \tau(t)]$: the topology of the two-sweep sum–product schedule with affine-plus-tanh updates in place of the exact ones, and the best claim of the three to being able to *learn* what belief propagation computes. Table 8.2 sets the five arms side by side. All network arms are trained with Adam at batch size 64, gradient clipping at 1, a shared learning-rate grid and a shared checkpoint ladder from 500 to 64,000 steps.

Table 8.2: The five arms as a ladder of supplied structure. “Order” means the architecture knows the sites are ordered, “sharing” that one rule is reused across sites, “radius” the furthest a prediction can depend on evidence. Only EM–BP is told the generating family, and only EM–BP is denied the clean trajectory. Parameter ranges are those swept in the screen; the selected window head ($r = 2$, $W = 64$) carries 4,801.

estimator	order	sharing	kernel form	radius	observes	free parameters
EM–BP, (6.1)	yes	yes	yes	L	x	24
window head, (8.6)	yes	yes	no	r	(a, x)	4.8k–21.4k
dilated stack, (8.7)	yes	yes	no	15	(a, x)	66.4k–264k
message passing, (8.8)	yes	yes	no	L	(a, x)	9.0k–34.3k
global MLP, (8.5)	no	no	no	L	(a, x)	25,248

8.3 Pointwise data efficiency

Against the unstructured denoiser, the structured estimator attains lower relative score error by a factor of 6.6–10.9 across the 8 training-set sizes (Table 8.3, Figure 8.1a), and is more accurate in all but 2 of the 1,520 scored (seed, size, level) cells. Every tunable quantity is chosen on a validation bundle disjoint from training and test, with the same privilege for both arms: the network selects its output head and training length, EM–BP its iteration count, and that choice coincides with the test-set optimum in 81.1% of cells for the network and 77.6% for EM–BP. If each arm’s error follows $\mathcal{E} \approx cN^{-\varrho}$ over the

tested range, then

$$R(N) = \frac{c_{\text{base}}}{c_{\text{EM}}} N^{\varrho_{\text{EM}} - \varrho_{\text{base}}}, \quad (8.9)$$

so the ratio grows precisely when the structured arm converges faster in data. The fitted finite-range slopes are $\varrho_{\text{EM}} = 0.332 \pm 0.013$ and $\varrho_{\text{base}} = 0.256 \pm 0.001$ (descriptive, not asymptotic), predicting that a 128-fold increase in data multiplies the ratio by 1.45; the measured ratio moves from 6.9 to 10.9, a factor of 1.58, non-monotonically. Neither slope is the parametric $1/2$: EM–BP’s mixture is misspecified against a Laplace innovation and its grid is finite, and the network’s function class is unrestricted. Equation (8.9) describes eight points and identifies no mechanism.

Table 8.3: Schedule-pooled relative score error against the deterministic grid-BP reference under the true kernel: squared error and reference score energy are summed over the twelve noise levels before the square root, per seed; 16 seeds $\pm$ one standard error. Both arms select their optimisation budget on a disjoint validation bundle. EM–BP is more accurate in 1,518 of 1,520 (seed, size, level) cells — a descriptive count, not 1,520 replications, since the twelve levels of a cell share a training set and a fitted model and the inferential unit stays the 16 paired seeds. The ratio column is the paired per-seed ratio of the two pooled errors, averaged over seeds, with a percentile bootstrap over those ratios beside it (seeds resampled, not cells) — close to, but not identical with, the quotient of the two columns beside it; Appendix A reports the same cells under five alternative aggregations. Protocol, cell exclusions and the budget calibration are in Appendix A.

sequences N	network	EM–BP (24 free)	ratio	95% CI
32	0.6024 ± 0.0046	0.0896 ± 0.0043	6.9 ± 0.3	[6.4, 7.5]
64	0.5025 ± 0.0042	0.0841 ± 0.0066	6.6 ± 0.5	[5.6, 7.6]
128	0.4238 ± 0.0024	0.0572 ± 0.0037	7.9 ± 0.5	[7.0, 8.8]
256	0.3494 ± 0.0017	0.0449 ± 0.0027	8.2 ± 0.4	[7.3, 9.0]
512	0.2891 ± 0.0010	0.0375 ± 0.0022	8.2 ± 0.5	[7.2, 9.2]
1024	0.2417 ± 0.0010	0.0294 ± 0.0025	9.0 ± 0.7	[7.7, 10.4]
2048	0.2033 ± 0.0007	0.0258 ± 0.0019	8.6 ± 0.7	[7.4, 9.9]
4096	0.1790 ± 0.0007	0.0180 ± 0.0015	10.9 ± 0.8	[9.4, 12.5]

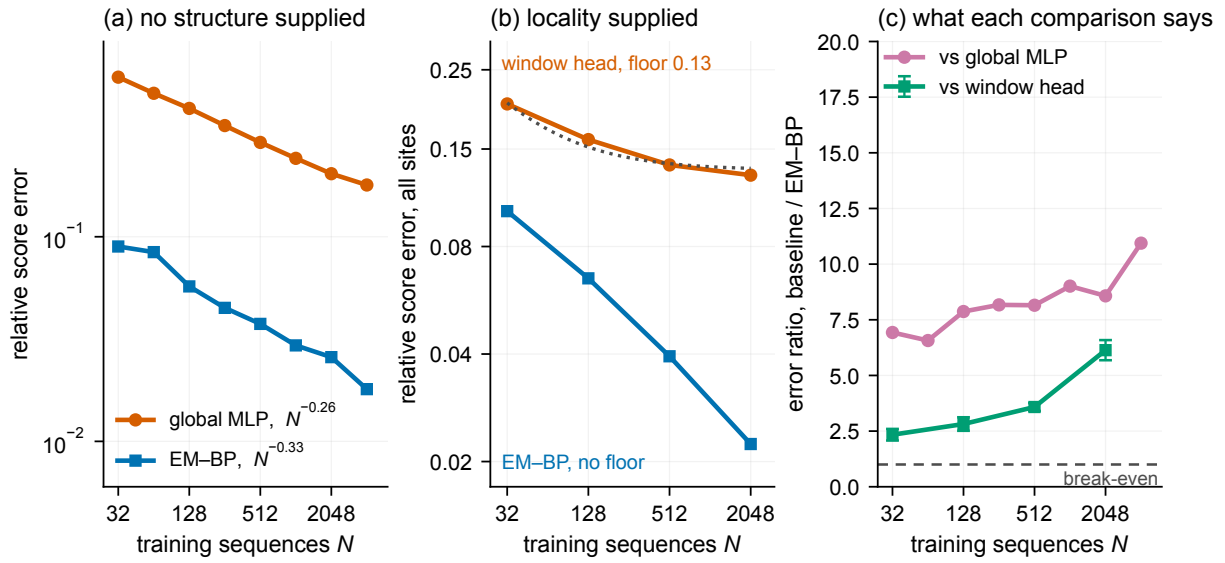


Figure 8.1: (a) Headline protocol: schedule-pooled relative score error (8.2) against training-set size for EM-BP and a denoiser given no structure, 16 seeds, with fitted finite-range slopes. (b) All-site protocol: the same for EM-BP and the screened window head, which is given locality and weight sharing; the dotted curve is the floor fit (8.14), whose fitted floor 0.13 sits on the last measured point. Read that as diminishing returns over four sizes, not a resolved floor: a floorless power law $N^{-0.11}$ fits the same points with a residual only 1.8 times larger. EM-BP’s curve is still falling steeply. (c) The two error ratios, $6.6\text{--}10.9\times$ above and the $2.3\text{--}6.1\times$ that survives once locality and sharing are supplied below; the gap is how much of the headline is not attributable specifically to the Markov model.

8.4 How much of the margin is the architecture?

8.4.1 The bulk posterior in closed form

How wide a window does a score head need? The variance-matched Gaussian chain answers before any network is trained, and the answer is a design calculation for the Laplace chain, whose second-order structure it shares. By Proposition 3.3 the posterior precision is $J = Q_0 + (e^{-2t}/\Delta_t)I$, the channel entering the diagonal alone, so in the bulk of a long chain J is tridiagonal Toeplitz,

$$J = \text{tridiag}(\beta, J_d, \beta), \quad J_d = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2}{\sigma_\eta^2}, \quad \beta = -\frac{\alpha}{\sigma_\eta^2}, \quad (8.10)$$

with only J_d depending on t .

Proposition 8.1 (Bulk covariance of the chain posterior). *In the interior of a long chain,*

$$(J^{-1})_{k, k+d} = V q^{|d|}, \quad q = \frac{J_d - \sqrt{J_d^2 - 4\beta^2}}{2|\beta|} \in (0, 1), \quad V = (J_d^2 - 4\beta^2)^{-1/2}, \quad (8.11)$$

the square root being real for every $|\alpha| < 1$ and $t > 0$.

Proof. Seek $G = J^{-1}$ in the form $G_{i, i+d} = V q^{|d|}$. For interior i and $d \geq 1$ the relation $\sum_j J_{ij} G_{jk} = \delta_{ik}$ reads $\beta q^2 + J_d q + \beta = 0$, the same quadratic for every d , whose roots multiply to one, so exactly one lies inside the unit disc. The $d = 0$ relation, $J_d V + 2\beta V q = 1$, gives V . Reality needs $J_d > 2|\beta|$, and $J_d - 2|\beta| = e^{-2t}/\Delta_t + (1 - |\alpha|)^2/\sigma_\eta^2 > 0$. $\square$

The formula agrees with a direct inverse at $L = 32$, $\alpha = 0.85$ to 6×10^{-15} at $t = 0.05$ and 2.7×10^{-4} at $t = 3$, the growth being the boundaries leaking into the bulk once the correlation length is an appreciable fraction of the chain; its limits are $q \rightarrow 0$ as $t \rightarrow 0$ and $q \rightarrow |\alpha|$, $V \rightarrow 1$ as $t \rightarrow \infty$. Because $m = (e^{-t}/\Delta_t) J^{-1} x$ is linear in the observation, the influence of distant evidence is exactly geometric,

$$\frac{\partial m_k}{\partial x_{k \pm d}} = \frac{e^{-t}}{\Delta_t} V q^d = \frac{e^{-t} V}{\Delta_t} e^{-d/\xi}, \quad \xi := \frac{1}{\log(1/q)}, \quad (8.12)$$

a pure exponential in the separation with a single length $\xi(\alpha, t)$, exact in the infinite-chain bulk: the object a score head has to represent, and the closed form behind the geometric decay measured in Section 4.4 (RQ4). The valid risk anchor is the Bayes-optimal radius- r estimator of the design model, the windowed posterior mean of (4.10), whose population relative score error pooled over the schedule with the same weights is

$$\mathcal{W}(r) = \left(\frac{\sum_t \text{tr} [(M_r(t) - Q_t) \Sigma_t (M_r(t) - Q_t)^\top]}{\sum_t \text{tr} Q_t} \right)^{1/2}, \quad (8.13)$$

the best any radius- r estimator can do on the covariance-matched Gaussian model. The experiment is Laplace, whose local oracle is a different quantity and no result ranks the Bayes risks of two priors sharing a covariance, so it is measured by window-restricted grid BP (Table 8.4): the Laplace oracle is the larger wherever the difference

is resolved, and the two are indistinguishable by $r = 8$.

Table 8.4: The exact Gaussian windowed-oracle error (8.13) and the Laplace chain’s own, pooled over the schedule at $\alpha = 0.85$, $L = 32$; the Laplace column is Monte Carlo over 1024 chains, and at $r = 2$ exceeds the Gaussian one by 14.4 bootstrap standard errors.

r	0	1	2	4	8
Gaussian $\mathcal{W}(r)$	0.657	0.202	0.089	0.029	0.007
Laplace $\mathcal{W}_L(r)$	0.664	0.215	0.094	0.030	0.007

Table 8.5: The decay rate q of (8.11), the correlation length ξ of (8.12), and the closed-form weight of each level in the pooled risk, at $\alpha = 0.85$ (eight of twelve levels). The two right-hand columns pull in opposite directions: evidence travels furthest exactly where the pooled score error is least sensitive.

t	$q(\alpha, t)$	$\xi = 1/\log(1/q)$	weight w_t (%)
0.0500	0.203	0.63	16.7
0.0725	0.259	0.74	14.4
0.1053	0.323	0.88	12.2
0.1527	0.392	1.07	10.2
0.3215	0.541	1.63	7.1
0.6769	0.689	2.69	5.4
1.4250	0.806	4.65	4.8
3.0000	0.848	6.06	4.7

The prediction follows without reference to any fit. The pooled risk is concentrated where $\xi \lesssim 1$, so a small radius is enough where it matters: at $t = 0.1527$ a radius of 4 discards $q^5 = 0.9\%$ of the influence, and the oracle error has fallen to 0.09 at $r = 2$ and 0.03 at $r = 4$; what remains beyond such radii comes almost entirely from the high-noise levels, whose weight is a few percent. *Prediction:* the best window radius is small, a few sites; the risk should rise again well beyond it, the receptive field then buying a quantity smaller than the estimation error it costs; and the selected head should sit well above $\mathcal{W}(r)$ at its radius, because at small radii the window is no longer what limits a learned head.

8.4.2 The screen and the confirmatory comparison

Three candidates were screened on 4 development seeds, disjoint from the 16 used for the comparison, each over its own hyperparameter grid, a shared learning-rate grid, both output heads and a shared checkpoint ladder, 32 configurations in all, scored on the bulk region. The window head won at radius 2 and width 64, the bidirectional message passing network next (0.148 against 0.187); Table 8.6 and Figure 8.2 report the screen in full. The architectures separate by parameter count far more cleanly than by risk, and the two arms built to propagate further occupy the expensive half of the axis without buying anything there, so the outcome is not one arm being starved of capacity.

Table 8.6: The architecture screen that selected Table 8.7’s baseline. Each candidate architecture was run over its own hyperparameter grid on 4 development seeds (disjoint from the confirmatory seeds), at $N \in \{32, 2048\}$, with a shared learning-rate grid, both output parameterisations, and a shared checkpoint ladder; the reported risk is the schedule-pooled relative score error, pooled over the development seeds and sizes on the interior (bulk) slice used for selection — the same estimand and accumulation as the selector inside the confirmatory job. $\star$ marks the winner carried into the confirmatory run. The dilated stack carries an order of magnitude more parameters than the winner and finishes last, so the outcome is not an artefact of one arm being given more capacity than another.

architecture	configurations	parameters	selection risk
weight-shared window head $\star$	16	4.8k–21.4k	0.148
dilated convolutional stack	8	66.4k–264k	0.226
bidirectional message passing	8	9.0k–34.3k	0.187

Both halves of the prediction hold. The radius sweep at width 64 gives **0.148**, 0.149, 0.185, 0.214 at $r = 2, 4, 8, 16$: minimised at 2, in a near-tie with 4, and rising steadily beyond. And the selected head sits well clear of the local oracle at its own radius, 0.148 against the experiment’s own $\mathcal{W}_L(2) = 0.094$ and the Gaussian anchor’s 0.089; the first difference is what remains to be won *inside* the radius-2 information set, learning, capacity and optimisation together, and it says the receptive field is not the binding constraint at the selected radius. This is a model-selection fact under this search space, protocol and development seeds, not evidence that long-range propagation is useless here.

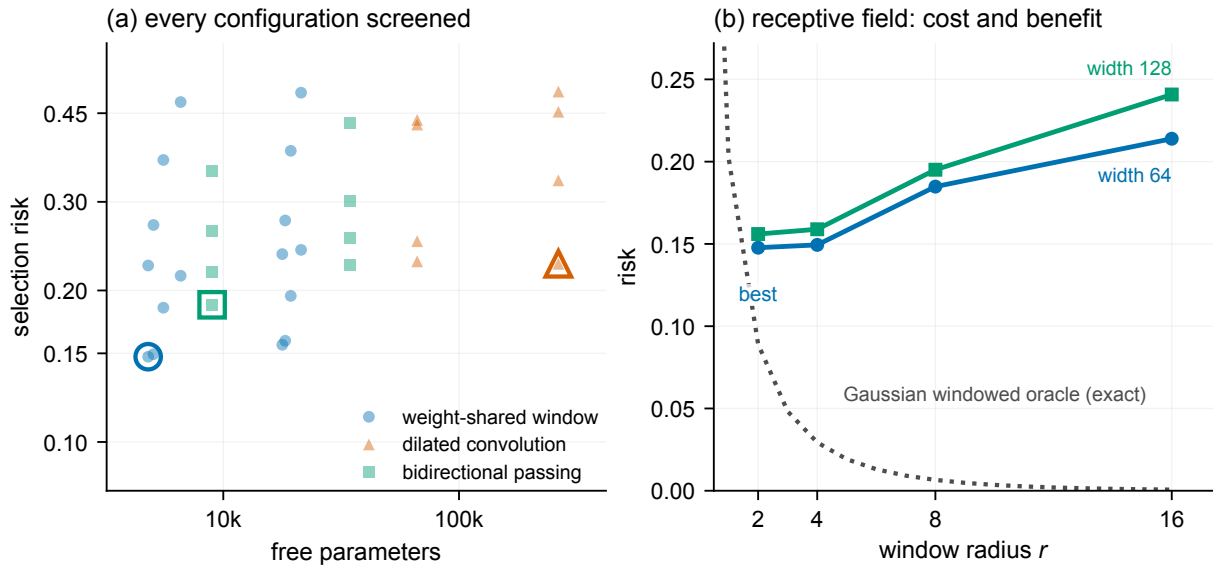


Figure 8.2: (a) Every configuration of the screen, scored on the bulk region, one marker per architecture and the outlined marker each architecture's best; the dilated stack sits an order of magnitude to the right and never reaches the window head's risk. (b) Risk against window radius at both widths, with the exact Gaussian windowed-oracle error (8.13) on the same axis: minimised at $r = 2$, nearly tied with $r = 4$, and rising beyond, as Section 8.4.1 predicts, while the oracle keeps falling.

Against the screened baseline, at the headline protocol throughout, with every site scored, the baseline's architecture and radius fixed on separate seeds and each arm's checkpoint chosen on a disjoint validation bundle, the surviving margin is $2.3\text{--}6.1\times$ (Table 8.7, Figure 8.1b,c). The dotted curve in panel (b) is the two-parameter floor fit

$$\mathcal{E}(N)^2 = \mathcal{E}_\infty^2 + \frac{c}{N}, \quad (8.14)$$

descriptive, as a four-point fit must be. EM–BP is more accurate in every one of the 192 scored cells, a count over 16 seeds, 4 sizes and 3 scoring regions that share fitted models, so the inferential unit behind each row is the 16 paired seeds; under 4 alternative aggregations none falls below $1.98\times$ at any size (Appendix A). A large fraction of the headline margin therefore does not survive supplying locality and weight sharing. The comparison does not decompose causally, since architecture, sharing, capacity and optimisation all change when the baseline is swapped; what it establishes is that the

generic-MLP ratio overstates what can be attributed specifically to the Markov model, by roughly a factor of three.

Table 8.7: EM–BP against the screened structure-aware baseline, at the headline protocol throughout. The baseline is a weight-shared window head whose architecture and hyperparameters were fixed by the screen of Table 8.6 on development seeds; only the checkpoint is chosen here, on a validation bundle disjoint from test, and EM–BP’s iteration count is chosen the same way. Risks are all-site and averaged over the 16 seeds; the ratio is $\text{risk}(\text{window})/\text{risk}(\text{EM–BP})$ formed within each seed and then averaged, $\pm$ one s.e. *All* scores every site, and is the headline; *centre* is a single-site diagnostic. A third region, the predeclared interior slice, is scored in the stored output and used for architecture selection but not reported here.

N	risk, all sites		ratio	
	EM–BP	window	all sites	centre site
32	0.1005	0.2006	2.34 ± 0.26	2.04 ± 0.24
128	0.0652	0.1594	2.82 ± 0.30	2.35 ± 0.26
512	0.0394	0.1353	3.58 ± 0.18	3.25 ± 0.18
2048	0.0224	0.1267	6.14 ± 0.45	5.43 ± 0.39

8.5 How much mixture capacity the estimator needs

The estimator’s one nonparametric dial is the number of mixture components. The capacity sweep refits $C \in \{1, 2, 4, 8, 16\}$ on identical data, paired within seed against the single-Gaussian innovation $C = 1$, at $N \in \{128, 512\}$, and scores held-out log-evidence (Table 8.8, Figure 8.3). Under this protocol no capacity buys anything: every paired mean is negative, the $N = 128$ deficit at the largest capacity is *resolved* against zero (-4.4×10^{-4} , CI $[-9.2 \times 10^{-4}, -1.1 \times 10^{-4}]$), and only two to four seeds in sixteen ever prefer a mixture. The sweep is not a clean statement about capacity in the abstract: above $C = 4$ every cell stops at the 1200-iteration cap rather than at tolerance, and the share of fits whose narrowest component falls below the grid’s resolution floor climbs with C , so a statistical disadvantage of large mixtures is entangled with an optimisation and a discretisation effect. The defensible conclusion is protocol-scoped: under this grid, M-step, iteration cap and validation rule, capacity beyond a single Gaussian in-

novation bought nothing, and Chapter 7 says why more capacity is also slower. The finding bears on the headline, which runs at $C = 8$ because the protocol fixed it in advance: a single Gaussian innovation with a fitted α is, by Theorem 5.2, exactly the covariance-matched linear estimator, so within the range tested the margin over the network arms is earned by the Markov factorisation and its covariance, not by the mixture’s capacity to represent a non-Gaussian shape.

Table 8.8: Held-out log-evidence per edge at each mixture capacity, paired within seed against the single-Gaussian innovation $C = 1$ on identical training and evaluation data, 16 seeds, each fit run to an iteration cap placed at its measured per-edge-gain plateau (the tolerance stop never fires there, so the cap is the effective stopping rule). Positive would mean the mixture buys something. No capacity does, at either size, and the sign count is the more honest summary of the two columns: the mean is dominated by a few large-magnitude seeds while the direction is consistent across most of them. The last column counts cells, over both sizes, whose narrowest fitted component falls below the grid’s resolution floor of 2 spacings — a share that climbs with C , and a reason not to read the table as a pure statement about capacity. Means and standard errors are in units of 10^{-2} .

C	$N = 128$			$N = 512$			
	mean	s.e.	seeds better	mean	s.e.	seeds better	under-resolved
2	−1.58	0.68	4/16	−1.29	0.67	3/16	4/32
4	−2.35	1.25	3/16	−1.28	0.66	2/16	8/32
8	−2.30	1.18	3/16	−1.04	0.49	2/16	10/32
16	−2.35	1.18	3/16	−1.30	0.65	2/16	11/32

8.6 Where the Markov assumption stops paying

Two controlled violations of the Markov assumption are measured against an exact reference. A *shared global latent* of strength β adds one standard normal draw per sequence to every site,

$$a = \frac{y + \beta g}{\sqrt{1 + \beta^2}}, \quad y \sim \text{chain prior}, \quad g \sim \mathcal{N}(0, 1), \quad (8.15)$$

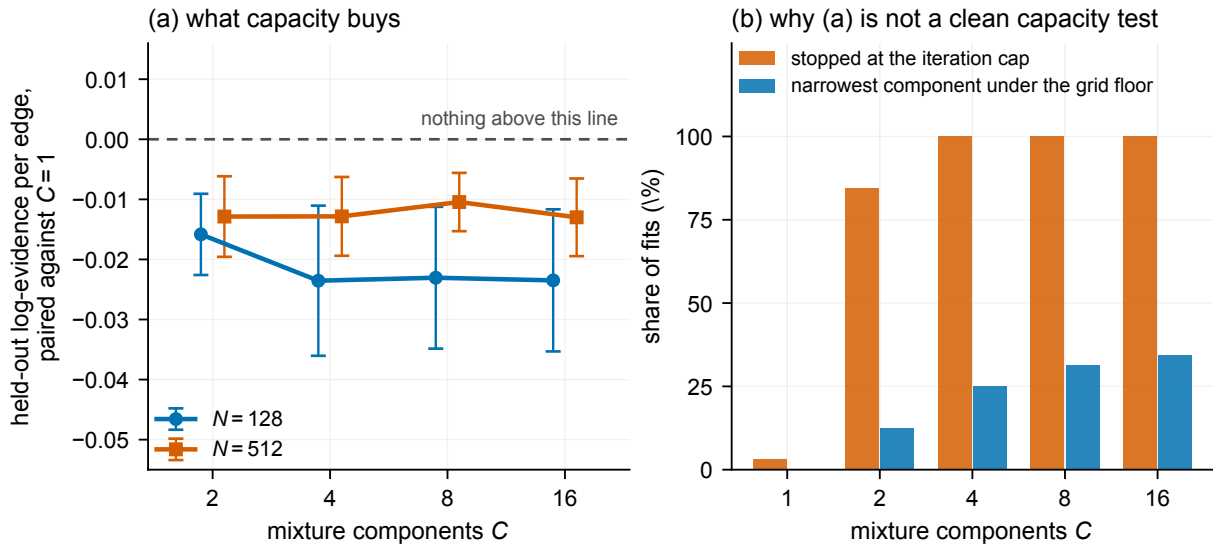


Figure 8.3: (a) Held-out log-evidence per edge, paired within seed against $C = 1$: no tested capacity improves on a single Gaussian innovation at either size. (b) Why (a) is not a clean capacity test: the share of fits stopped at the iteration cap, and with a narrowest component under the resolution floor, both climb steeply with C .

variance preserving, Markov only at $\beta = 0$, with covariance $\Sigma_\beta = (\Sigma_0 + \beta^2 \mathbf{1}\mathbf{1}^\top)/(1 + \beta^2)$, so that a Laplace and a Gaussian version differ only beyond second moments; at $\beta = 1$ half the marginal variance of every site is one shared constant. *Long-range precision coupling* of strength γ adds weak couplings directly to the precision at every separation of two or more,

$$Q_\gamma = Q_{\text{AR}} + \gamma Q_{\text{long}}, \quad [Q_{\text{long}}]_{ij} = -e^{-|i-j|/\ell} \quad (|i-j| \geq 2), \quad [Q_{\text{long}}]_{ii} = 1.05 \sum_{j \neq i} e^{-|i-j|/\ell}, \quad (8.16)$$

with $\ell = 8$, diagonally dominant by construction and rescaled to unit diagonal. The two mechanisms are not two amounts of the same thing.

Proposition 8.2 (The global latent perturbs the chain in one direction). *With $Q_{\text{AR}} = \Sigma_0^{-1}$ and $Q_\beta = \Sigma_\beta^{-1}$,*

$$Q_\beta = (1 + \beta^2) Q_{\text{AR}} - c_\beta u u^\top, \quad u = Q_{\text{AR}} \mathbf{1}, \quad c_\beta = \frac{(1 + \beta^2) \beta^2}{1 + \beta^2 \mathbf{1}^\top Q_{\text{AR}} \mathbf{1}}, \quad (8.17)$$

so the departure from a rescaled chain precision has rank one, and the lag-one corre-

lation of (8.15), hence that of its best first-order chain, is $\alpha_{\text{CL}}(\beta) = (\alpha + \beta^2)/(1 + \beta^2)$.

Proof. Sherman–Morrison applied to Σ_β gives (8.17). The covariance of (8.15) at separation $d \geq 1$ is $(\alpha^d + \beta^2)/(1 + \beta^2)$ with unit variance, and the KL projection of a Gaussian onto a fixed tree matches that tree’s node and edge marginals (Chow and Liu, 1968), so on a path the fitted chain reproduces the true adjacent correlation. $\square$

Numerically the correction has one nonzero eigenvalue at every β tested, with eigenvector agreeing with $u/\|u\|$ to six decimals, and u is, up to boundary terms, the constant vector; the correction γQ_{long} is dense, with numerical rank L . One direction against thirty is the structural difference, and a chain that may reparameterise a single scalar can partly trade against the first but not the second: under (8.15) the fitted $\hat{\alpha}$ climbs from 0.848 to 0.939 at $\beta = 1$, tracking α_{CL} to within 0.014, whereas under (8.16) the same gap is 7 times larger (Table 8.9). A single number makes the difference quantitative, the per-site KL divergence between the covariance-matched centred Gaussians of the contaminated law and of its best first-order chain,

$$\delta = \frac{1}{L} D_{\text{KL}}(\mathcal{N}(0, \Sigma) \parallel \mathcal{N}(0, \Sigma_{\text{CL}})) = \frac{1}{2L} \left[\text{tr}(\Sigma_{\text{CL}}^{-1} \Sigma) - L + \log \frac{\det \Sigma_{\text{CL}}}{\det \Sigma} \right], \quad (8.18)$$

closed-form for both mechanisms and involving no fit. For the Gaussian arms it is the exact KL to the best Gaussian chain; for the Laplace arms it is a covariance proxy, blind to everything beyond second moments, which is exactly the information the chain estimator’s design geometry uses. In nats per site it reads 4.6×10^{-6} , 1.5×10^{-4} , 1.4×10^{-3} and 6.5×10^{-3} at $\beta = 0.1, 0.25, 0.5, 1$, and 1.1×10^{-2} , 2.2×10^{-2} , 4.0×10^{-2} and 6.0×10^{-2} at $\gamma = 0.05, 0.1, 0.2, 0.4$: a prior in which half of every site’s variance is shared is closer to the chain family than one with the weakest long-range coupling tested. Table 8.9 and Figure 8.4 report the measured advantage beside it.

Table 8.9: Two controlled violations of the Markov assumption, each evaluated against the appropriate reference: the Gaussian rows against an exact linear-algebra reference for the contaminated model, the Laplace rank-one rows against the validated grid-BP reference. The first two rows of each block are the baseline’s relative score error divided by EM–BP’s, averaged over the noise schedule, so a value above 1 means the structured estimator wins and a bold value means it loses. The third row is the fitted correlation minus the Chow–Liu correlation of the true contaminated law: the distance between the chain that was fitted and the best chain there is. The fourth is EM–BP’s own relative score error at its worst noise level, which says how much of a change in the ratio is the estimator degrading rather than the baseline improving.

	global latent, rank-one, strength β				
	0.00	0.10	0.25	0.50	1.00
Gaussian, vs CNN	16.41	14.84	6.68	3.21	2.16
Gaussian, vs MLP	29.34	22.48	11.28	4.00	1.67
Laplace, vs CNN	11.14	10.92	6.21	2.96	2.09
Laplace, vs MLP	16.51	17.64	10.13	3.67	1.62
Gaussian: $\hat{\alpha} - \alpha_{\text{CL}}$	−0.0025	−0.0008	+0.0008	+0.0051	+0.0144
Gaussian: EM–BP error, worst t	0.019	0.025	0.055	0.134	0.167
Laplace: EM–BP error, worst t	0.022	0.022	0.047	0.132	0.183

	long-range precision coupling, strength γ				
	0.00	0.05	0.10	0.20	0.40
Gaussian, vs CNN	16.07	1.24	0.97	0.88	0.75
Gaussian, vs MLP	27.84	1.66	1.19	0.84	0.70
Gaussian: $\hat{\alpha} - \alpha_{\text{CL}}$	+0.0018	+0.0376	+0.0557	+0.0806	+0.0976
Gaussian: EM–BP error, worst t	0.016	0.298	0.402	0.490	0.579

Read against strength, the mechanisms differ in kind. At $\beta = 1$ the Gaussian arm is still ahead at every noise level against both baselines, worst ratio 1.06, and the Laplace arm is still ahead against the convolutional baseline, reaching break-even against the fully connected one only at the largest diffusion time. Under long-range coupling the advantage against the convolutional baseline is gone by $\gamma = 0.1$, having survived $\gamma = 0.05$, by $\gamma = 0.2$ the estimator loses to both baselines, and at $\gamma = 0.4$ its own relative

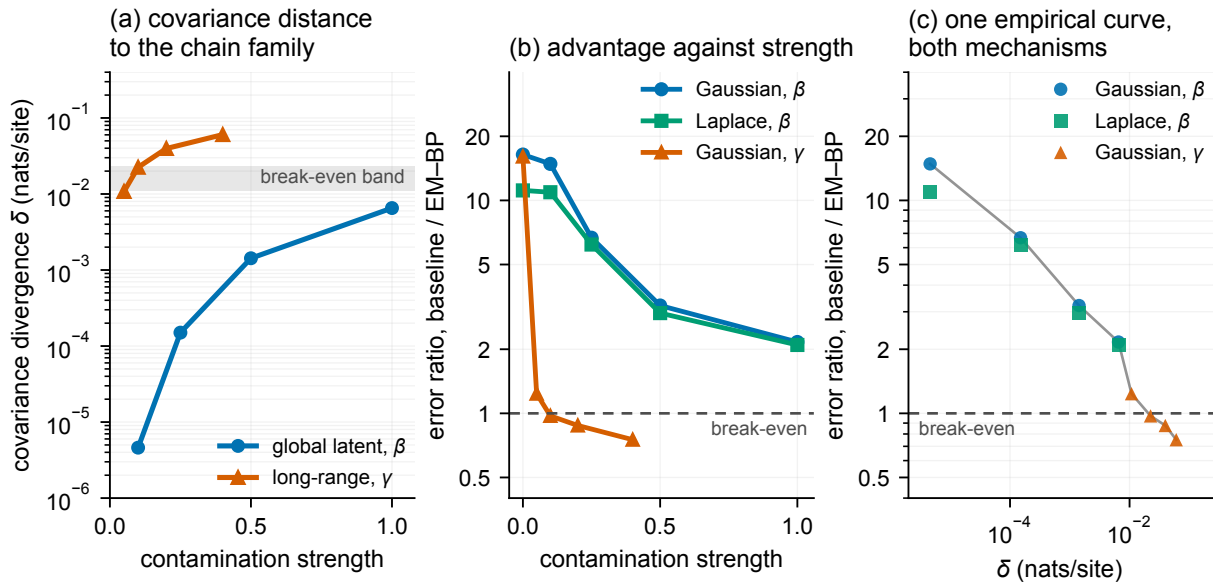


Figure 8.4: (a) The closed-form distance (8.18) to the chain family; the mechanisms reach the shaded break-even band at strengths differing by more than an order of magnitude. (b) The measured advantage against strength: under a global latent it decays smoothly and does not reach the line even where half the marginal variance is shared; under long-range coupling it crosses within the first tested strength. The long-range panel is Gaussian only, the contaminated Laplace model having no closed-form reference. (c) The same nine points against δ : the mechanisms interleave and fall on one monotone curve, crossing break-even between the strengths bracketed in (8.19). All curves fall because the estimator degrades, not because the baselines improve (Table 8.9).

score error reaches 58%. Read against δ , the nine contaminated cells interleave and the advantage is monotone in δ across both mechanisms, over four orders of magnitude in δ and a factor of twenty in the ratio, against both baselines and for both innovation laws. The crossing is reported as the bracket the design resolves,

$$\delta^* \in [1.1, 2.2] \times 10^{-2} \text{ nats/site (vs CNN)}, \quad [2.2, 4.0] \times 10^{-2} \text{ nats/site (vs MLP)}; \quad (8.19)$$

each condition is a single run with no seed replication, so no interval can be attached to a crossing and no point estimate is quoted. The collapse onto one curve is an empirical regularity of the two violation paths constructed here, not a sufficiency theorem: even between Gaussians equal KL does not imply equal score geometry, since the divergence is a spectral aggregate while the score error of a chain approximation depends

on where the precision error points, and adjacent strengths differ by roughly a factor of two, which is the resolution of the statement.

8.7 Why the shape coordinate is the expensive one

Chapter 7 attributed the slow convergence of the innovation shape to the fourth cumulant being damped by the channel while the second is preserved. That is a claim about the noised likelihood, and it can be checked against Fisher information independently of any optimiser. For $X_t = e^{-t}A + \sqrt{\Delta_t}\varepsilon$, cumulants add under independent sums and the Gaussian contributes only to the second, so $\kappa_2(X_t) = 1$, $\kappa_4(X_t) = e^{-4t}\kappa_4(A)$ and $\kappa_{11}(X_{t,i}, X_{t,i+1}) = e^{-2t}\alpha$: the autoregressive coefficient, identified through the lag-one cumulant, is attenuated by e^{-2t} , a shape parameter moving the fourth cumulant at fixed variance by e^{-4t} . Reading information as squared sensitivity over a sampling variance that is t -independent to leading order because $\kappa_2(X_t) \equiv 1$ pins the scale, a delta-method heuristic rather than a likelihood calculation,

$$\boxed{I_\alpha(t) \asymp e^{-4t}, \quad I_p(t) \asymp e^{-8t}, \quad \frac{I_p(t)}{I_\alpha(t)} \asymp e^{-4t},} \quad (8.20)$$

and the third statement is the robust one, a contrast from which common omitted factors cancel.

The prediction is tested on generalised Gaussian innovations $f_p(\eta) \propto \exp(-|\eta/s|^p)$ with s fixed by $\text{Var}(\eta) = \sigma_\eta^2$ at every p , so that p is a genuine shape coordinate ($p = 1$ Laplace, $p = 2$ Gaussian). For $\theta = (\alpha, \sigma_\eta^2, p)$, Fisher's identity (6.7) supplies the exact per-sequence score of the marginal log-likelihood from one forward-backward pass, the information matrix is estimated in outer-product form over $D = 512$ sequences of $L = 32$ sites, and the quantity reported is the *efficient* information $I_{p,(\alpha,\sigma_\eta^2)} = 1/(J^{-1})_{pp}$, what remains about p once the nuisance coordinates must be estimated too. Two caveats travel with every number: an outer-product matrix is the Fisher information only at the true parameter of a correctly specified model and is here a D -sample estimate;

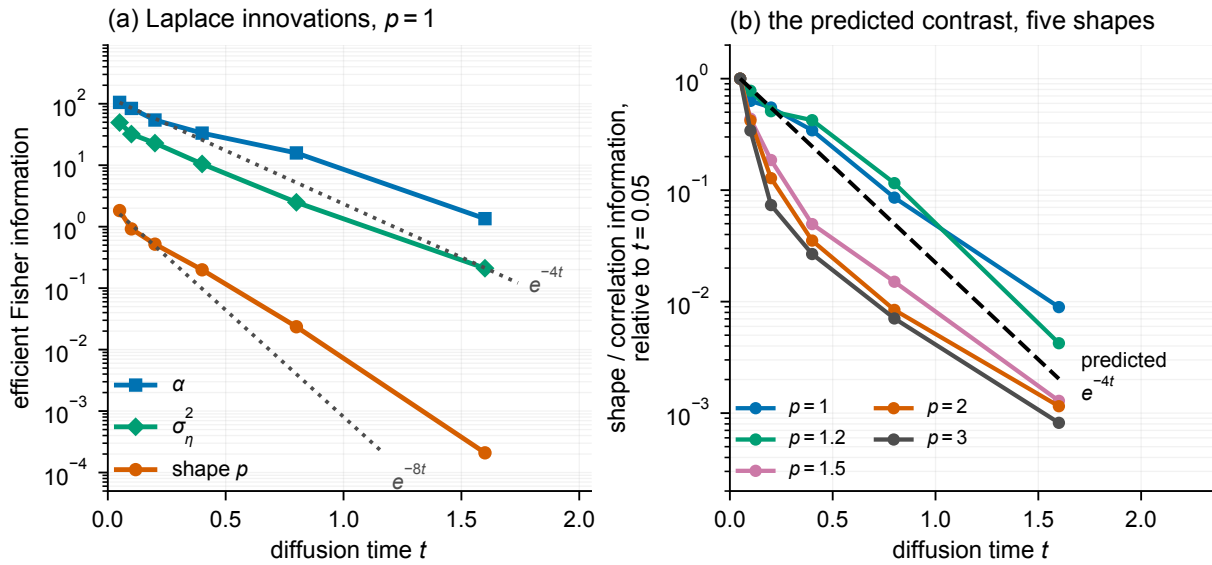


Figure 8.5: (a) Efficient information for each coordinate against diffusion time at Laplace innovations, with the predicted rates (8.20) as dotted guides; on a linear time axis an exponential is a straight line. (b) The contrast I_p/I_α for all five shape exponents, normalised at $t = 0.05$, against e^{-4t} .

and the reported quantity is an inverse whose meaning degrades with the conditioning of J , benign where the information is large ($\kappa(J) = 61$ at $t = 0.05$, median 2776) and failing where it vanishes ($1.3e + 07$ in the worst cell), so the far tail says the information has gone, not how much remains. Figure 8.5 and Table 8.10 report the result: both measured decays are shallower than predicted by a common offset, and the contrast, which the offset cancels, brackets the prediction with a mean fitted log-slope of -3.72 against -4 and a geometric-mean information loss of 465 against a predicted 493.

Table 8.10: How much efficient Fisher information about each coordinate the channel destroys between $t = 0.05$ and $t = 1.6$, and the contrast (8.20) predicts; entries are ratios of outer-product estimates at $D = 512$, so only the order is meaningful. The geometric mean of the last column is 465 against a predicted $e^{4 \times 1.55} = 493$.

true p	excess kurtosis	loss, α	loss, σ_η^2	loss, p	contrast	loss _{p} /loss _{α}
1.0	+3.00	78×	235×	8.7×10^3		112
1.2	+1.74	87×	392×	2.0×10^4		236
1.5	+0.76	83×	459×	6.4×10^4		776
2.0	0.00	75×	501×	6.5×10^4		867
3.0	−0.58	56×	522×	6.8×10^4		1222

8.8 What the chapter establishes

Four findings, each at the strength its evidence supports. Supplying the chain is worth 6.6–10.9× lower pooled error over 8 sizes against no structure, under the schedule-pooled paired estimand with both arms budget-selected on validation. Against a baseline given locality and weight sharing the margin is 2.3–6.1×, EM–BP ahead in every scored cell, for one screened architecture and four sizes, with no causal split of the gap; most of the headline factor does not survive, which bounds how much of it can be attributed specifically to the Markov model. No tested mixture capacity beats a single Gaussian innovation under this grid, M-step, iteration cap and validation rule. And across the tested violations the advantage ordered monotonically with a covariance scalar computable before any fitting, with break-even bracketed between adjacent strengths, an empirical regularity of nine cells and two mechanisms, useful as a screening variable and not established as a decision rule. Throughout, the coordinate on which every non-Gaussian claim depends is simultaneously the slowest to converge and the poorest in information per observation, which is why the budget disclosures in this chapter are part of its results.

Chapter 9

Discussion and Conclusions

9.1 Answers to the research questions

For the stationary Gaussian AR(1) chain under coordinatewise OU noise the joint score is linear and known in closed form, $S(x, t) = -Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$; its couplings fill one diagonal per order in t from a tridiagonal start and return to the identity at rate e^{-2t} (RQ1). The same score is computed by sum-product on the posterior factor graph, a tree because every observation enters through a unary factor, and on the Gaussian chain the recursion is the Kalman filter followed by RTS smoothing (RQ2). Off the Gaussian case the recursion stays exact but its messages do not close, and Theorem 5.2 identifies Gaussian projection as the LMMSE estimator of the covariance-matched model, so the measured gap to a validated grid reference (median relative score error 0.27 at $t = 0.05$, below 10^{-2} by $t \approx 1$) is the price of discarding structure beyond second moments (RQ3). In the bulk the influence of evidence at distance d decays exactly as $q(\alpha, t)^d$, and one hop of message passing removes 85% of the isolated-frame error at $t = 0.05$ (RQ4). Fisher’s identity turns one forward-backward pass into the exact likelihood gradient, so an unknown transition law is recovered from noised sequences by EM with nothing differentiated through the recursion, the innovation shape settling at a median of 229 updates against 80 for the correlation (RQ5). At an equal number of clean trajectories the recovered structure is worth $6.6\text{--}10.9\times$ lower schedule-pooled relative score error than an unstructured network and $2.3\text{--}6.1\times$ against one given locality and weight sharing, under a stated aggregation and with the arms

asymmetric in supplied information by design (RQ6).

9.2 What is new, under which assumptions, and what comes next

Three things are new: the exact structure of a chain diffusion score, with chain BP identified with the Kalman–RTS recursion; an exact reading of what Gaussian closure computes, which makes the measured Laplace gap the price of a model class rather than of an implementation; and transition learning by exact chain inference inside EM, compared at equal data, with its boundaries mapped: no gain from added mixture capacity under the implemented grid and cap, partial absorption of a rank-one Markov violation against failure under long-range coupling near strength one tenth, and a two-rate convergence law.

The exact results concern one-dimensional states, linear AR(1) dynamics and a known channel, with the locality laws proved for the Gaussian bulk only; the non-Gaussian results are numerical, grid BP being a reference only within its validated regime; and the learning results are statistical, with the headline factors moving by tens of percent under defensible alternative aggregations (Appendix A), every capacity cell above $C = 4$ stopping at an iteration cap, the fitted convergence slopes being contraction factors rather than Jacobian eigenvalues, and the misspecification scalar being an exact KL only in the Gaussian arms. Whether generative fidelity tracks pointwise accuracy within one estimator, and whether trained networks on real sequence data inherit any of these properties, are untested here.

Four directions follow: rerunning the reverse-generation comparison on convergence-selected fits rather than a fixed iteration count; a grid-by-capacity design to separate resolution effects from overfitting; a paired design across violation strengths and types; and, beyond the chain, the reading of U-Nets as belief propagation (Section 2.5), with tree exactness replaced by an approximation whose cost must be measured as Chapter 5 measures the Gaussian closure.

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Finally, I thank my family, to whom this thesis is dedicated, and my friends, who heard a great deal about Markov chains with patience.

Appendix A

Reproducibility

A.1 Build identifier and figure-to-code map

Every number in the thesis is produced by a script from frozen experiment outputs under `research/outputs/frozen/`; none is typed by hand (Table A.1). The quoted experiment numbers are defined once as macros in `sections/` and cited by name, so tables, figures and text share one aggregation rule by construction. The build manifest records the state this document was made from: repository revision `46aaca051084`, generated inputs `f62c391c8e5d7e5b` over 19 files, figures `fcadc4d1db009f43` over 18 files, engine Tectonic 0.16.9. It fixes the source side only; the provenance of the frozen outputs is disclosed below. The companion compendium, distributed with the sources, is the extended research record of this thesis: the same results with every derivation written out.

A.2 Provenance and protocol facts

The closed-form and quadrature studies are deterministic and run on a laptop; the estimator comparisons ran under SLURM from the scripts in `hpc/`, and four provenance routes appear, in decreasing order of strength. *Certified*: `exp_31` (screen and confirmatory comparison) and `exp_32` (capacity) were deployed by a script that refuses a dirty tree, ships a `git` archive of the commit and records its digest. *Legacy digest*, *empty intersection*: `exp_07`, the headline comparison, predates that script and is citable be-

Table A.1: Every computed figure and generated table, and the script under `tools/` (thesis) or `research/tools/` (research package) that builds it, with the frozen experiment it reads where it reads one. Figure 4.1 is reproduced from the literature.

figure / table	built by
Figure 2.1	<code>make_thesis_figures</code> , closed forms
Figures 3.2, 3.3, 3.4	<code>fig_ch05</code> , closed forms
Figure 4.3	<code>fig_ch06</code> , closed forms
Figures 5.1, 5.2, 5.3	<code>fig_ch07</code> , quadrature; exp. 02 (frozen)
Figures 6.1, 6.2, 6.3	<code>fig_ch08</code> , exps. 01, 18, 06 (frozen)
Figure 7.1	<code>fig_ch09</code> , exp. 27 (frozen)
Figures 8.1, 8.2, 8.3, 8.4, 8.5	<code>fig_ch10</code> , exps. 07, 31, 32, 21, 22 (frozen)
Table 8.3	<code>make_tab_efficiency</code> , exp. 07
Tables 8.6, 8.7, A.2	<code>make_tab_screening</code> , <code>make_tab_structured</code> , <code>make_aggregation_structured</code> , exp. 31
Tables 8.8, 8.9, 7.2	<code>make_tab_capacity</code> (exp. 32), <code>make_tab_nonmarkov</code> (exp. 21), <code>make_convergence_numbers</code> (exps. 27, 06)
Laplace oracle of Table 8.4	<code>make_laplace_oracle</code> , quadrature

cause the intersection of each run’s recorded dirty-file list with its own import closure is empty, a check recomputed on every regeneration; the one row where it was not, $N = 8192$, is withdrawn. *Recorded configuration only:* the validation and diagnostic runs (exp_01, 02, 06, 18, 22, 27) carry the resolved configuration and seeds but an empty revision field, and the claims resting on them are stated as measured curves and orderings. *Certified after rerun:* exp_21, behind Table 8.9, is a clean-deployment rerun whose Gaussian arm reproduces the run it replaces on every quoted number.

Both arms of the headline comparison select their budget on a disjoint validation bundle, agreeing with the test-set optimum in 81.1% and 77.6% of cells; rerunning $N = 2048$ at the raised budget of the $N = 4096$ row moves its pooled ratio by $+0.53 \pm 0.20$, in the direction that makes the capped table conservative for the structured arm. The resolution gate excludes 16 of 1,536 cells whose narrowest fitted component is under two grid cells wide; including them changes no ratio by more than 0.03. The screen ran 32 configurations over 4 development seeds disjoint from the confirmatory seeds, with

learning rates 0.001, 0.003, both output heads and a checkpoint ladder from 500 to 64000 steps. The settling times of Section 7.3 come from the 800-iteration shard of `exp_27`, the only run length at which no configuration settles at its own cap. The environment is Python ≥ 3.10 with NumPy, SciPy, pandas and Matplotlib, float64 throughout, the neural baselines in pure NumPy with an explicit random generator so that training is reproducible from the recorded seed; these are version floors rather than a lock, so byte-level reproduction is not claimed, only determinism given the seeds on a fixed installation. The document compiles with one run of `tectonic main.tex`, and `check.sh` reports unresolved references, undefined citations, duplicate labels and overfull boxes.

A.3 How the reported ratios were aggregated

A ratio is a nonlinear function of two means, so it depends on the order in which averaging is done, and both reported ratios were recomputed under alternative aggregations of the same frozen cells, refitting nothing. For the headline (Table 8.3) the reported estimand is the schedule-pooled paired per-seed ratio of (8.2); the mean of per-(seed, level) ratios, which earlier drafts reported, is systematically the largest alternative ($7.0\text{--}17.6\times$ against the pooled $6.6\text{--}10.9\times$), because a per-level ratio at a level with little reference energy can be very large while contributing little. The direction is the same under every aggregation, every paired bootstrap interval over seeds sits well clear of one, and no aggregation produces a reversal at any size; resolved by noise level, the per-level ratio peaks near $t \approx 0.5$, and EM–BP is beaten outright in 2 of 1,520 cells. For the structure-aware comparison the reported estimand is not uniformly the largest, but the floor across the four aggregations is $1.98\times$ with a spread of 0.48 (Table A.2), so the reading of a low single-digit multiple at the smallest training set growing to roughly six at the largest survives every one of them.

Table A.2: Table 8.7's all-site ratio under four aggregations of the same cells, with the paired bootstrap interval over seeds. (1) is what the table reports. The conclusion does not depend on the choice: no estimand at any size falls below $1.98\times$, and the largest departure from the reported value is 0.48.

N	(1) per-seed	(2) ratio of means	(3) geometric	(4) median	(5) 95% CI on (1)
32	2.34	2.00	2.15	1.98	[1.88, 2.86]
128	2.82	2.45	2.61	2.62	[2.29, 3.44]
512	3.58	3.43	3.51	3.59	[3.25, 3.91]
2048	6.14	5.66	5.89	5.93	[5.30, 7.02]